

Article title

Influence of source area size on modelling urban heat island intensity and land cover relationships across Local Climate Zones

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Declaration of competing interests

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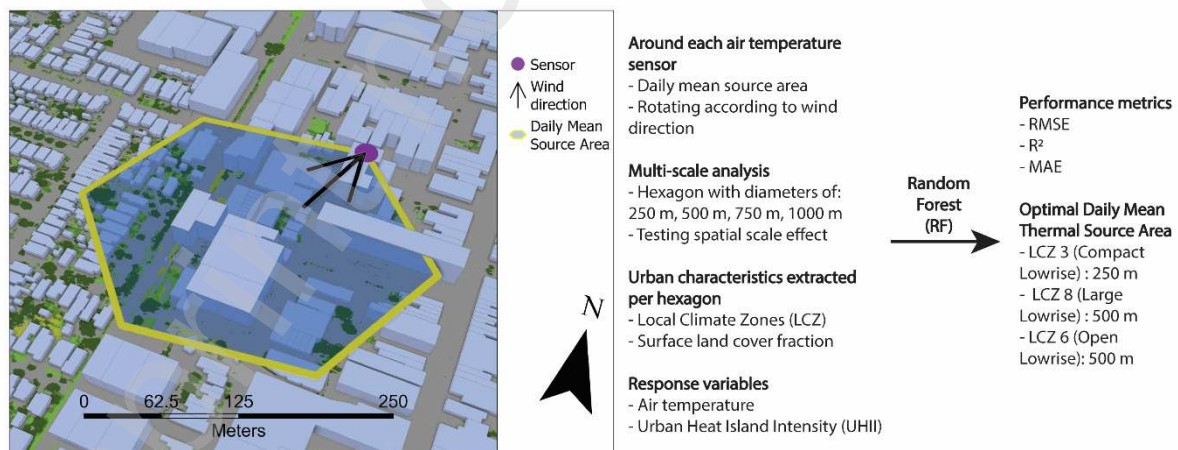
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Abstract

Understanding how specific urban morphology features influence the Urban Heat Island (UHI) effect requires robust characterisation of Local Climate Zones (LCZs) and appropriate definition of thermal source areas. This study developed separate daytime and nighttime Random Forest (RF) models to determine the spatial scale that best represents the thermal source area based on surface cover fractions. The response variable, UHI intensity (UHII), was derived from air temperature data collected from Citizen Weather Stations (CWS) and professionally operated weather stations (PRWS) under stable atmospheric conditions (low cloud cover, low wind speed, low relative humidity, and no precipitation within the preceding 24 hours). Predictor variables included percentages of impervious surfaces, trees, shrubs, wet and dry grass, water, and bare soil. LCZs were generated using the LCZ Generator. Four hexagon diameters (250 m, 500 m, 750 m, and 1000 m) were tested in 12 directional sectors (0° – 330° at 30° intervals), aligned with prevailing wind directions to account for turbulent heat transport toward sensors. Results show that daytime UHII is generally lower than nighttime UHII. Nighttime patterns are similar for LCZ 3 (compact low-rise) and LCZ 8 (large low-rise), whereas LCZ 6 (open low-rise) exhibits rapid heating and cooling. The highest UHII values occurred during consecutive hot days, indicating heat accumulation. Model performance suggests a 500 m diameter as optimal for LCZs 6 and 8 ($R^2 = 0.814$; 0.859), while LCZ 3 performs best at 250 m ($R^2 = 0.617$), reflecting differences in morphology and heat storage capacity.

Keywords: Urban Heat Island Intensity, Local Climate Zones, Surface cover fraction, Thermal source area, Air temperature, Random Forest



1 Introduction

Buildings, pavement, and other infrastructure in cities absorb and hold heat, causing urban areas to be warmer than nearby rural areas. This is called the Urban Heat Island (UHI) effect (Howard, 1833). Human activities that change land use and land cover alter urban features such as building density and sky view factor, affect land surface properties such as emissivity and albedo, and ultimately impact the surface energy balance (Li et al., 2015; Oke et al., 2017). The magnitude of the UHI is usually stronger at night, when the temperature difference between urban and rural areas is at its highest. This happens because urban materials absorb heat during the day and release it slowly overnight, keeping urban temperatures higher (Oke, 1982).

To accurately measure UHI, it's important to distinguish between UHIs quantified using air temperature data and those using land surface temperature (LST) data. Measurements based on air temperature reflect the urban canopy layer (UCL), which is the lowest part of the atmosphere, extending from the ground up to the height of trees or buildings (Arnfield, 2003; Stewart, 2011). In the UCL, Oke (1988) has defined the underlying causes for the development of the UHI based on the energy budget equation, which comprises: (1) absorption of shortwave and longwave radiation, (2) reduced evapotranspiration due to limited vegetation cover, (3) heat released from human activities, and (4) the obstruction of airflow caused by urban structures.

Tobler's First law of Geography says that "everything is related, but closer things are more connected than distant ones" (Tobler, 1970). This is especially true in local scale urban climate studies, where the mix of buildings, roads, vegetation, soil, and water strongly affects air temperature. The physical characteristics, surface material and volume of these features affect local climate by modifying airflow, heat transport, radiation balances, albedo, moisture availability, and ground heating/cooling potential. This concept is modelled as Local Climate Zones (LCZ), which are defined "as regions of uniform surface cover, structure, material, and human activity that span hundreds of meters to several kilometres in horizontal scale" (Stewart & Oke, 2012, p. 1884). This distance range corresponds to the thermal source area, which is the total area influencing the air temperature at a specific location.

The thermal source area for a temperature measurement is the total surface area "seen" by the sensor (sometimes called its "footprint" or "circle of influence") (Stewart & Oke, 2012). Many researchers have avoided exploring the concept of the thermal source area within LCZs (Núñez Peiró et al., 2019) because classical source-area theory was developed for sensors located in the internal sublayer, where assumptions of horizontal homogeneity and well-developed turbulence apply (Schmid, 2002). In contrast, most urban temperature measurements are made within the UCL, where airflow is strongly influenced by buildings and vegetation (Oke, 2025), causing these assumptions to break down. Consequently, thermal source areas within LCZs become difficult to define in a physically rigorous manner, discouraging their explicit use in urban climate studies.

Instead, researchers often rely on the approximation provided by Stewart and Oke (2012) or selecting a cell size that aligns with their local climate analysis, such as the width of a city block or the pixel resolution of a specific remote sensing sensor. For example, previous studies

have used 250 x 250 m square grid cells as the basic spatial unit for representing LCZs (Chen & Hu, 2022; Dimitrov et al., 2021; Liu & Shi, 2020). However, the thermal climate across LCZ boundaries is spatially continuous, as the screen-level (1–2 m above ground) air temperature of one zone gradually merges with that of adjacent zones (Stewart & Oke, 2012). Consequently, the sharp edges of pixels or the irregular shapes and sizes of city blocks do not accurately represent distinct climatic units.

Existing thermal source area models in the literature use a variety of methodological approaches, including analytical dispersion solutions, Lagrangian Stochastic (LS) models, and computational fluid dynamics (CFD) methods (Kljun et al., 2002; Schmid, 2002). While these physics-based methods involve thermodynamic processes and may be grounded in the physical basis of the UHI (Grimmond et al., 2010), they also require detailed information about the materials, geometries, and local weather to accurately represent the built environment (Adilkhanova et al., 2022). Additionally, they require specialised expertise and resources to conduct the modelling (Acosta et al., 2024; Mirzaei, 2015).

Data-driven approaches have shown strong potential in UHI studies without relying on heat exchange models or advanced numerical simulations (Acosta et al., 2021; Adilkhanova et al., 2022; Oukawa et al., 2022). To effectively develop such models for the UCL, accurate ground-based temperature measurements are essential. Because LCZ boundaries arise from the advection of heat and moisture across areas with different surface structures and land cover (Stewart & Oke, 2012), understanding how surfaces influence urban temperatures is crucial. High spatial and temporal resolution air temperature data are crucial for a data-driven definition of thermal source areas. However, collecting such data is difficult, making it hard to compare areas with different surface roughness, LCZ classes, and land cover types (Liu et al., 2017; Romero Rodríguez et al., 2023). With the growing availability of dense networks of Citizen Weather Stations (CWS) and high-resolution land cover data, there is an opportunity to develop empirical methods to estimate the size of thermal source areas under real-world conditions.

This research aims to investigate the influence of surface land cover on UHII across different LCZ classes, and to develop a data-driven approach for estimating thermal source area size using temperature observations from PRWS and CWS. In doing so, four different sizes of hexagonal thermal source areas were evaluated, with each hexagon rotated around one vertex in 12 directions to account for wind orientation. The surface cover fraction within each rotated hexagon was then extracted to assess its influence on the observed UHII. Determining the thermal source area ideally relies on air temperature and quantifies the influence of surface cover fraction on temperature changes. However, to better capture the impact of urbanisation and the increase of impervious surfaces both horizontally and vertically, air temperature data were converted into UHII by comparing measurements to a reference station in a non-urbanised area. This approach shifts the focus from solely identifying thermal source areas to assessing how added impervious surfaces drive changes in air temperature.

2 Materials and methods

This section outlines the methodology used to assess the influence of urban surfaces on local-scale heat. It combines air temperature measurements, the calculation of UHII, and machine learning models to estimate the effective thermal source areas around monitoring sites.

2.1 Study area

Five northern suburbs of Melbourne were selected including the Local Government Areas (LGAs) of Brimbank, Maribyrnong, Moonee Valley, Merri-bek, and Darebin. These LGAs are characterised by their increasing population density and ongoing urban development. Melbourne's northern suburbs feature predominantly flat topography. Using the airport and selected Bureau of Meteorology (BOM) sites as benchmarks, the elevation difference between the southern and northern parts is approximately 90 m, while the difference between the eastern and western parts is about 60 m (see **Table 1**). The LGAs are characterised mainly by LCZ 6 (Open Lowrise) and are bordered by LCZ 3 (Compact Lowrise) and LCZ 8 (Large Lowrise), particularly to the south near Port Phillip Bay and to the north. The surrounding natural environment includes LCZ D (Low Plants) and LCZ G (Water).

Table 1: List of PRWS from the Bureau of Meteorology

Weather station	Station code	Elevation (m)	Latitude	Longitude
Laverton RAAF	087031	20.1	-37.8565	144.7565
Melbourne Olympic Park	086338	7.5	-37.8255	144.9816
Essendon Airport	086038	78.4	-37.7276	144.9066
Melbourne Airport	086282	113.4	-37.6654	144.8322
Viewbank	086068	66.1	-37.7408	145.0972

Melbourne's Köppen-Geiger climate classification is "Cfb", which means it has a temperate oceanic climate with warm summers and consistent rainfall throughout the year (Kottek et al., 2006). On average, Melbourne gets about 531 mm of rain annually. The mean maximum temperature is 13.4°C in July (winter) and 25.8°C in January (summer) (Keay & Simmonds, 2005). Melbourne is influenced by maritime air from Port Phillip Bay, with the coastline located less than 50 km from the study area. **Figure 1** shows the map of the study area, including the locations of BOM stations across metropolitan Melbourne and the LGAs.

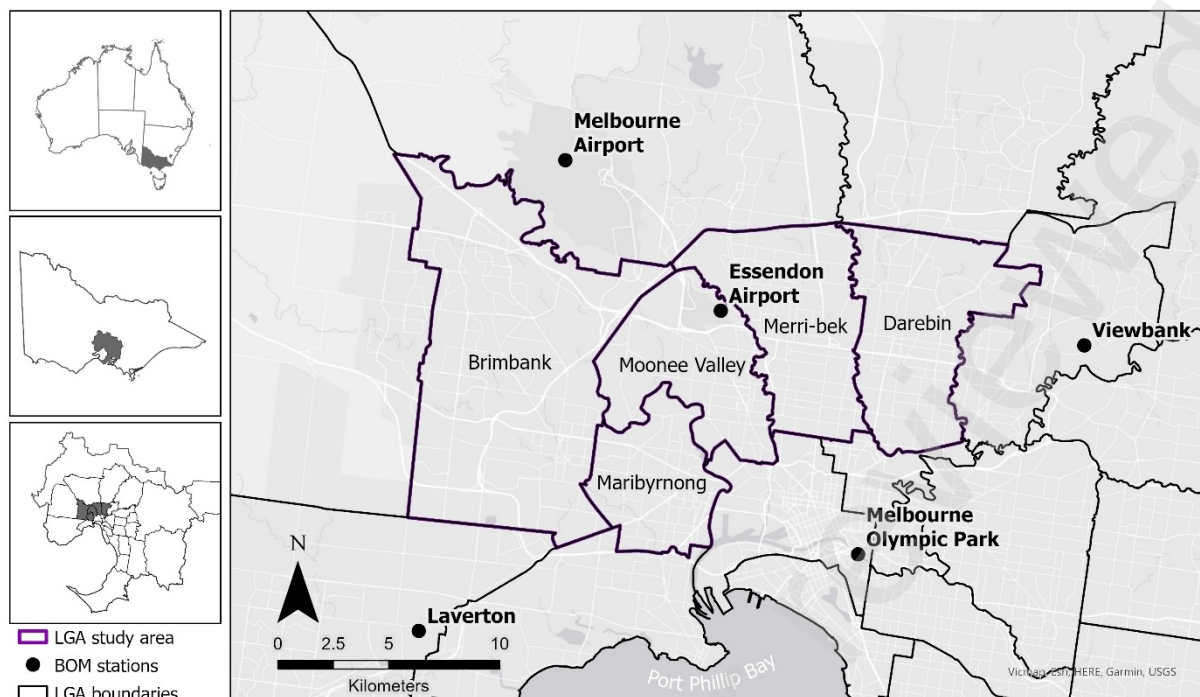


Figure 1: Study area and location of Bureau of Meteorology (BOM) stations in Melbourne, Australia.

2.2 Selection of days with minimal weather influence on UHI

The UHII is known to be highly sensitive to local meteorological conditions such as cloud cover, wind speed, and relative humidity (Sundborg, 1951). These weather factors typically show negative correlations with UHII, particularly wind speed, followed by cloud cover and humidity levels, and outweigh the influence of surface land cover (Oukawa et al., 2022). As a result, identifying days with minimal meteorological interference is essential for assessing the land-surface-driven components of UHI.

Previous research has shown that calm, clear nights offer the most favourable conditions for observing strong UHII effects attributable to urban surface characteristics (Montávez et al., 2000). For example, Morris et al. (2001) found that UHII in Melbourne averaged 2–3°C on such nights, compared to less than 1°C under windy conditions. The presence of cloud cover reduces daytime solar heating and suppresses nighttime radiative cooling. Similarly, wind enhances atmospheric mixing, diminishing the thermal contrast between urban and rural areas.

Light winds are generally defined as speeds below 2 m/s (Al-Obaidi et al., 2021), which limit turbulent mixing. Clear skies enhance the sky heat sink at night, promoting stronger radiative cooling in rural zones. Lower relative humidity may also support clearer atmospheric conditions, contributing to stronger land-surface-driven UHI formation. Finally, days were selected if there was no precipitation in the preceding 24 h (Stewart et al., 2014).

The threshold values for key meteorological parameters used to identify days with strong land surface influence and limited atmospheric interference are summarised in **Table 2**.

Table 2: Threshold values for key weather parameters used in the analysis

Weather parameters	Threshold	Explanation
Daily mean percentage of cloud cover	25%	UHII is greatest when cloud cover < 10% (Morris et al., 2001)
Daily mean windspeed	5 m/s	Maximum mean wind speed used by (Oke, 1973)
Daily mean percentage of relative humidity	70%	The relative humidity is negatively correlated with the maximum UHII (Kim & Baik, 2002). The mean relative humidity between December 2022 and February 2023 was 70%.

To maximise solar radiation exposure, the analysis focused on summer days, specifically from 1 December 2022 to 28 February 2023. A diurnal cycle was defined from sunrise to sunrise the following day. Within this cycle:

- Daytime was defined by positive solar irradiance (sun above the horizon)
- Nighttime began when irradiance dropped to zero (sunset), indicating the absence of direct solar input.

After filtering down to days with minimal weather influence, a set of ten candidate days across December to January was identified. During this period, sunrise occurred consistently around 06:00 (± 15 minutes) and sunset around 20:30 (± 15 minutes). Days in February were excluded due to windy or cloudy conditions.

2.3 Monitoring sites

2.3.1 Data extraction at monitoring sites

Providing air temperature data with high spatial and temporal resolution remains a challenge in urban climate research as at least one of these conditions is often not met. The study area, along with a 5 km buffer, includes five PRWS managed by the Bureau of Meteorology (BoM). Although PRWS data undergo strict quality control and comply with standardised conditions, such as specific instrument housing, installation height, enclosure design, and site exposure, these setup requirements limit their ability to represent diverse urban environments. As a result, they often capture data only from open areas or a single type of LCZ (typically LCZ D - Low Vegetation) and fail to reflect the microclimates within built-up urban areas.

Therefore, in addition to data from PRWS, quality-controlled observations from CWS were used, sourced from two platforms: Netatmo and Wunderground (see **Figure 2**). Netatmo sensors have an accuracy of $\pm 0.3^\circ\text{C}$, while Wunderground stations may range between ± 0.5 and $\pm 1^\circ\text{C}$ (Lizana et al., 2023). To assess the longer-term consistency of CWS data, observations were collected over the summer period, from 1 December 2022 to 28 February 2023.

Netatmo data were provided by the Bochum Urban Climate Lab through the ENLIGHT (ENabling the analysis of Global urban HeaT) project. The project aims to develop a consistent and globally applicable methodology for UHI analysis using satellite Earth observation and

crowdsourced data (Ching et al., 2018). For Wunderground, hourly mean air temperature values were derived from archived 5-minute data available online.¹

Metadata such as geographical coordinates (longitude, latitude, elevation), hardware specifications, and software platforms were collected. Quality control of CWS data was performed using the “wisdom of the crowd” approach as described by Fenner et al. (2021). Further validation was conducted through kriging interpolation of CWS data, followed by extraction of predicted values at PRWS locations and comparison with observed values. This procedure and the detailed results of the data quality control process are provided in the **supplementary material**.

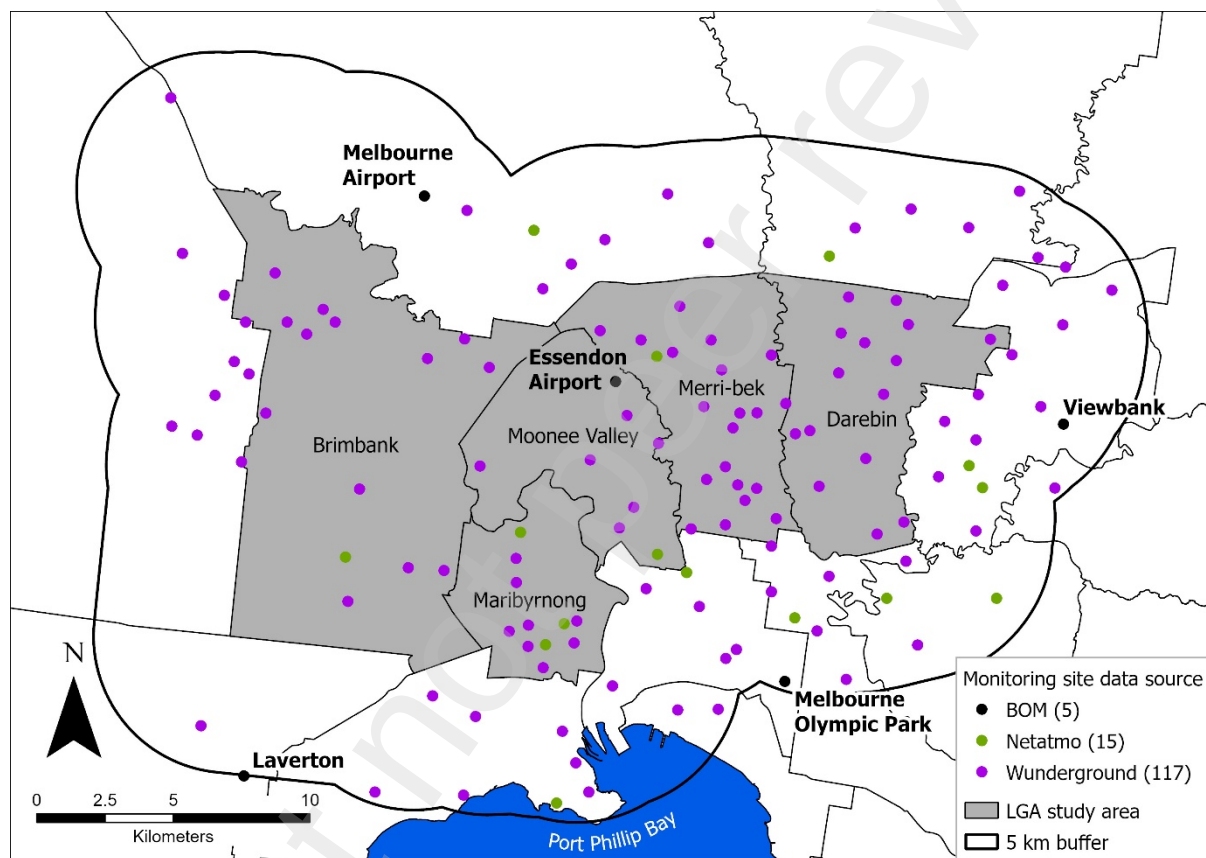


Figure 2: Location of monitoring sites

2.3.2 Local Climate Zone classification over the study area

The LCZ classification scheme developed by Stewart and Oke (2012) is applied in this study. Each LCZ represents a distinct combination of buildings, roads, vegetation, soil, rock, and water, arranged in varying proportions and configurations to form 17 standardised patterns. The classification was performed online using LCZ Generator, which is a platform developed by WUDAPT (World Urban Database and Access Portal Tools) that applies remote sensing techniques where the user defines training data (Demuzere et al., 2021). WUDAPT produces

¹ <http://www.wunderground.com/>

consistent LCZ maps for cities worldwide at a standard spatial resolution of 100 m (Bechtel et al., 2019; Ching et al., 2018; Demuzere et al., 2021; Wellinger et al., 2024). The classification itself is performed on the Google Earth Engine platform using a random forest classifier, in which pre-processed Landsat 8 and Sentinel 1/2 data input features are used (Demuzere et al., 2021). **Figure 3** shows the LCZ map generated using the LCZ Generator tool.

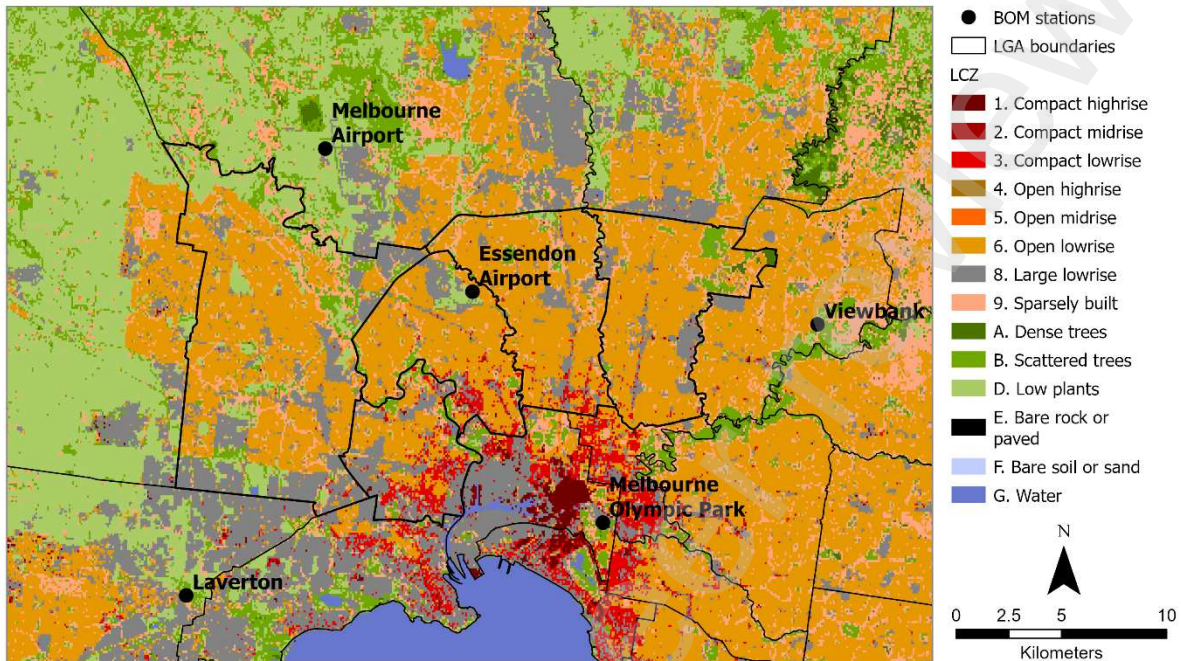


Figure 3: LCZ map generated using the LCZ Generator tool

2.3.3 Size and shape of the thermal source area

The size and shape of thermal source areas can change over time, and their theoretical form resembles an elliptical or circular shape (Stewart & Oke, 2012). Its location relative to the sensor depends on the wind direction. **Figure 4** shows hypothetical source areas and surface wind directions for a temperature sensor (1–2 m above ground) under prevailing southeasterly winds within the LCZ classification scheme developed by Stewart and Oke (2012).

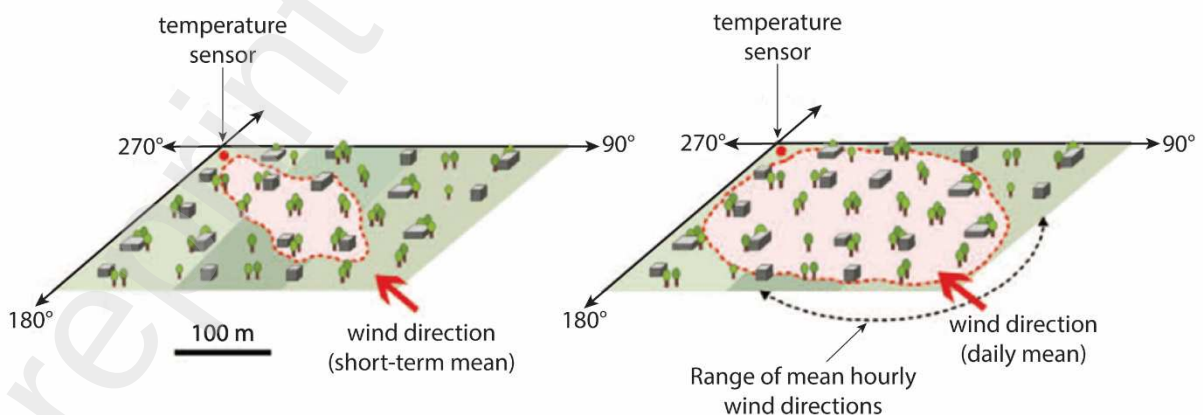


Figure 4: Illustration of the thermal source area in relation to the LCZ classification scheme. Left: Short-term source area (<1 hour). Right: Mean daily source area. Source: (Stewart & Oke, 2012)

In practice, there are three regular tessellations of a plane to represent the environment in ecological models, including climate-ecosystem interactions: triangles, squares and hexagons (Carr et al., 1992). In urban climatology, square grids are commonly used due to their simplicity, ease of implementation, and compatibility with remote sensing and GIS data. This is particularly evident in LCZ mapping and urban canopy models, where regular grids facilitate spatial analysis and model coupling. Regular grids, especially hexagonal ones with their uniform connectivity and radial symmetry (Birch et al., 2007), may represent thermal source areas more effectively than squares due to their ability to approximate circular shapes and support of isotropic flow representation. Here, four hexagon sizes (250 m, 500 m, 750 m, and 1000 m in diameter) were tested. These sizes correspond to the physical principle of the thermal source area, whose diameter typically ranges from 400 to 1000 m, reflecting the 200–500 m fetch required for air at screen height to adjust to the underlying surface conditions, which depend on surface roughness, urban form, and atmospheric stability (Oke, 1987; Stewart & Oke, 2012; Wieringa, 1993).

The thermal source area around each sensor was arranged according to twelve orientations, ranging from 0° to 330° at 30° intervals in polar coordinates. The placement is determined by the wind direction at each monitoring site on each day of observation. **Figure 5** illustrates the 12-direction wind rose along with an example arrangement of four hexagons of different diameters, with wind blowing approximately from the north-northwest toward the south-southeast (150° in polar coordinates).

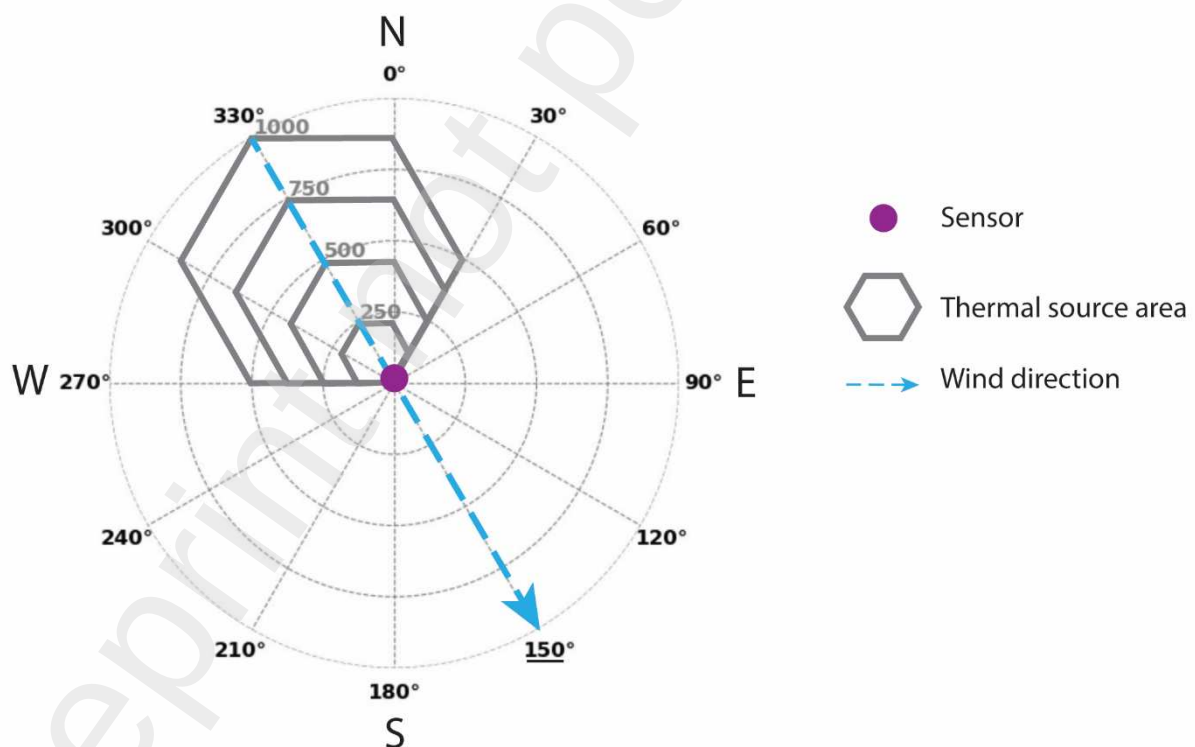


Figure 5: Wind rose and example layout of four hexagons of varying diameters. The wind direction is 150° in polar coordinates.

The daily mean wind direction at each monitoring site was derived from the interpolated (IDW: Inverse distance weight interpolation) wind direction data from the five PRWS (**Figure 6**).

IDW is a deterministic interpolation method where values are estimated as a weighted average of nearby observations, with weights inversely proportional to distance (Shepard, 1968). The PRWS anemometers are positioned 10 m above the ground (Holmes, 2022); the measured wind reflects conditions over open, unobstructed terrain upwind of the station, providing a stable representation of wind direction across the study area.

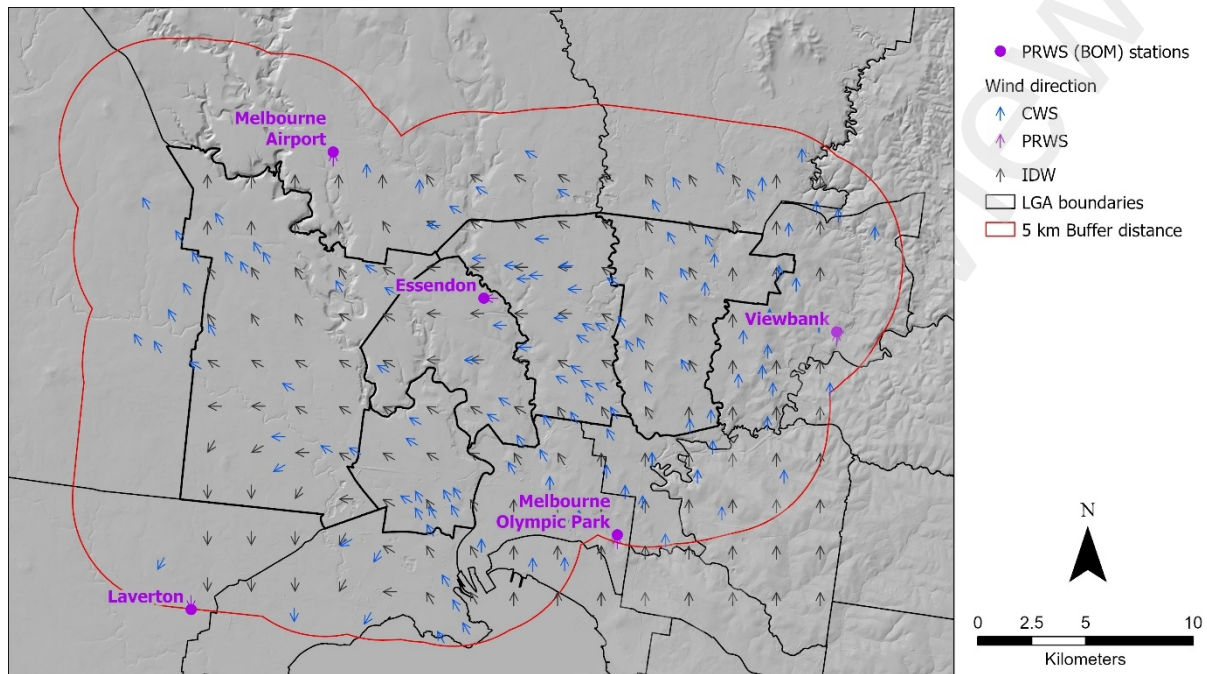


Figure 6: Mean wind direction on 3 December 2022, showing wind at the PRWS location, corrected wind directions at CWS locations, and the IDW-interpolated wind field over a hillshade topographic map.

2.3.4 Assigning Local Climate Zones to monitoring sites

Each hexagon was assigned an LCZ class using morphological filtering following Vavassori et al. (2023). The LCZ class of each hexagon was determined from the underlying 100 m square grid it overlapped. The LCZ classification around the monitoring site was examined using Google Street View imagery (**Figure 7**) to verify the local morphology. Historical imagery was set to December 2022, the closest available date to the weather data collection period in January 2023.

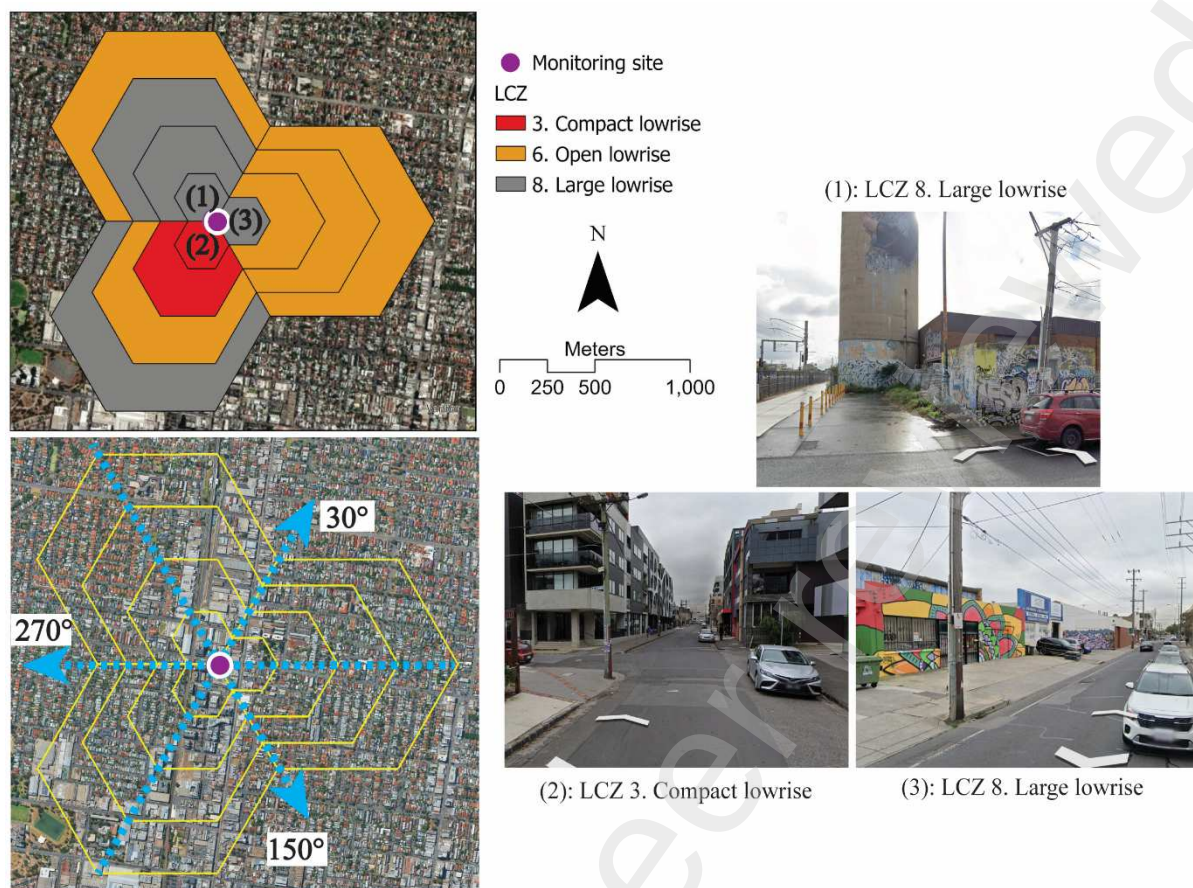


Figure 7: LCZ classes around a monitoring site in Merri-bek (Albion Street), showing three wind directions (Cyan dashed lines) and corresponding Google Street View imagery.

2.3.5 Monitoring sites and surface land cover fraction

Land cover data from 2018, with a spatial resolution of 0.2 meters, provided by the Commonwealth Scientific and Industrial Research Organisation (CSIRO), were used in this study (Hurley et al., 2019). The dataset includes six land cover classes: low infrastructure, high infrastructure, grasses (vegetation height < 0.5 m), shrubs (0.5–3 m), trees (height > 3 m), and water bodies. Low infrastructure includes roads and streets (asphalt/concrete), footpaths, parking lots and paved surfaces. High infrastructure includes low- and high-rise buildings, bridges, and elevated roads. Additionally, cloud-free PlanetScope images (3 m resolution) were used to identify dry grass areas that would typically appear green following adequate rainfall.

For each hexagon, the surface cover fractions were calculated and associated with the corresponding wind direction and hourly air temperature to assess their influence on UHII. The spatial distribution of land cover types within the study area is shown in **Figure 8**, with an inset illustrating three 250 m-diameter hexagons rotated around one vertex at 30° (wind blowing to the NNE), 150° (SSE), and 270° (W) to reflect different orientations.

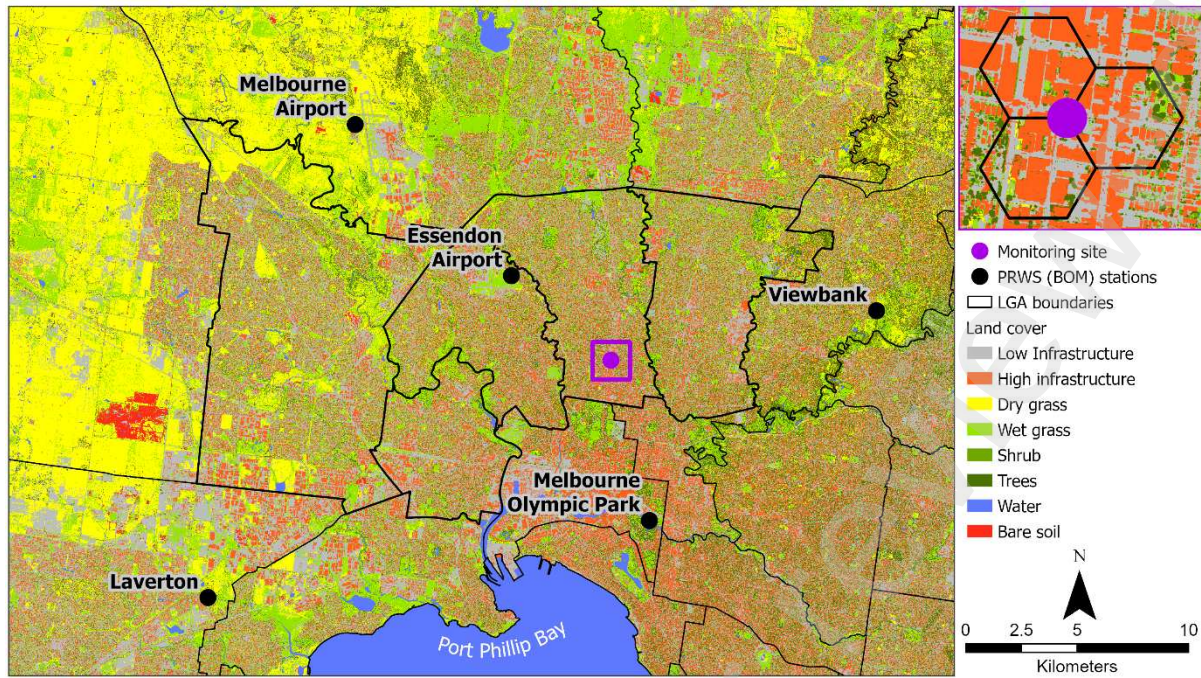


Figure 8: Land cover map of the study area with an inset showing three 250 m-diameter hexagons oriented according to different wind directions (W, NNE, and SSE) at one example monitoring site.

2.4 Definition of the rural area and calculation of urban heat island intensity (UHII)

Following the definition by Oke (1982), the UHII formula is:

$$UHII = \Delta T_{LCZx-r} = T_{LCZx} - T_r$$

Where T_{LCZx} is the average hourly temperature at monitoring sites with same LCZ type, and T_r the average hourly temperature at a rural station.

Defining a suitable rural reference station presents a significant challenge in UHI analysis. Undisturbed natural landscapes are rarely found near urban areas (Erell & Williamson, 2007; Runnalls & Oke, 2000). The peri-urban zone is often dominated by agricultural land or areas previously altered by human activity, which may not represent a truly rural thermal baseline. As a result, rural reference stations are often located at considerable distances from the urban core, introducing potential mesoscale climatic influences.

In the case of Melbourne, different urban climate studies have selected different stations as rural references. Morris et al. (2001) defined the rural temperature as the average of values recorded at Melbourne Airport, Laverton, and Moorabbin Airport. In contrast, Sachindra et al. (2016) selected only Laverton as the rural reference station, citing data availability and consistency, particularly due to missing records at the other stations during their study period. Most recently, Rogers et al. (2019) identified Melbourne Airport alone as the rural station. These variations highlight the lack of a consistent definition and the methodological challenges in selecting an appropriate rural benchmark for UHI studies. In this study, the latest definition has been adopted, with Melbourne Airport used as the rural reference station.

2.5 Construction and validation of a model of the relationship between UHII and surface land cover fraction by LCZ type and size

A Random Forest (RF) regression model was developed for the diurnal cycle, encompassing both daytime and nighttime, to investigate the spatiotemporal evolution of UHII and assess how urban morphology influences the size of the thermal source area. Studies have shown that RF performed well when predicting the air and surface temperatures from predictors such as land cover (Noi et al., 2017; Oukawa et al., 2022; Voelkel & Shandas, 2017).

RF is a non-parametric, nonlinear machine learning algorithm that employs a mixed-information approach by constructing and aggregating multiple decision trees to improve prediction accuracy and robustness (Breiman, 2001; Cutler et al., 2007). The two key parameters in an RF model are the number of trees and the number of predictor variables. Based on these, each tree in the RF model keeps splitting the data into increasingly smaller subsets until it can't be split any further. The final prediction is made by averaging the outcomes from all the trees, allowing the model to classify most datasets with high accuracy.

To evaluate the generalisability of the RF model, the following validation steps were applied:

- Split the dataset into 80% for training and 20% for testing
- Train the model using the training subset, with 100 trees and 300 repeats
- Predict the UHII using the testing data, and
- Calculate performance metrics including R^2 , Mean Absolute Error (MAE), and Root Mean Square Error (RMSE).

3 Results

3.1 Classification of monitoring sites by LCZ

Monitoring sites were assigned to LCZ classes based on the majority LCZ type within the surrounding 250 m hexagons around each weather station across all wind directions. **Table 3** presents the distribution of monitoring sites across LCZs for each data source. Most monitoring sites are from CWS. Among the PRWS, one site is classified as LCZ 8 (Melbourne Olympic Park), and two as LCZ D (Essendon Airport, and Laverton).

Table 3: Number of monitoring sites per LCZ by data source

LCZ	CWS	PRWS	Total
LCZ 3. Compact lowrise	14	0	14
LCZ 6. Open lowrise	97	0	97
LCZ 8. Large lowrise	15	1	16
LCZ D. Low plants	0	2	2
Total	126	3	129

3.2 Surface land cover fraction per LCZ at monitoring sites

To understand the spatial composition of land cover around monitoring sites, the surface land cover fractions were summarised by LCZ using the four hexagon diameters (250 m, 500 m, 750 m and 1000 m). **Figure 9** shows that LCZ 3 (Compact Low-rise) and LCZ 8 (Large Low-rise) are dominated by impervious surfaces which accounts for between 62.5% and 75% of the

hexagon's area. LCZ 6 has a similar proportion of low infrastructure as LCZ 3 and LCZ 8, but displays a more balanced land-cover composition, with a larger share of vegetated areas such as wet grass and tree canopy. LCZ D is predominantly covered by vegetation (over 80%), with dry and wet grass being the dominant land cover types.

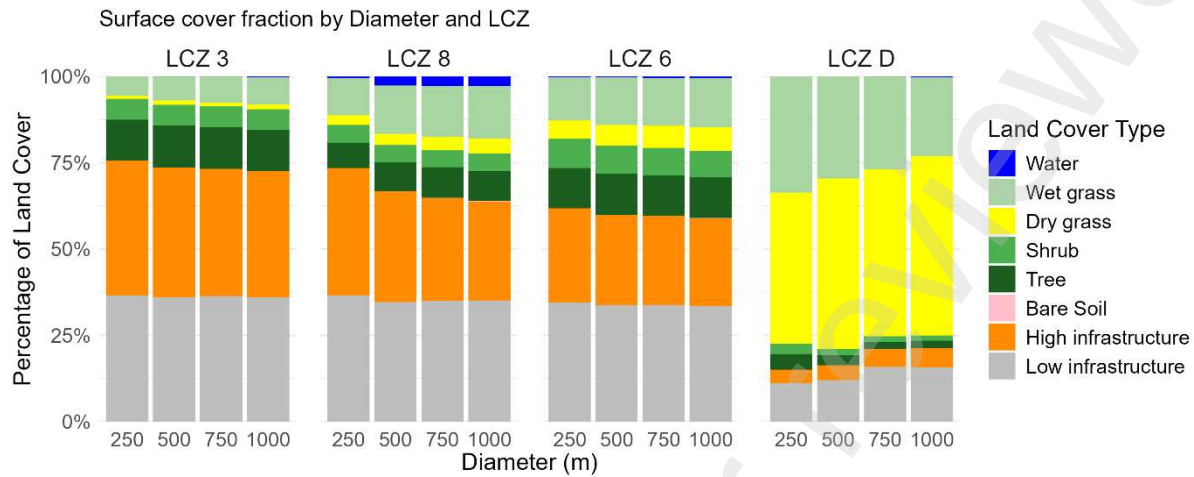


Figure 9: Mean surface land cover fraction by hexagon diameter and LCZ

3.3 Diurnal Variation of UHII: inter- and intra-class comparisons across LCZs

Figure 10 shows the diurnal distribution of mean UHII across LCZs 3, 6, and 8 monitoring stations, averaged over the twelve days of observation. The x-axis starts at 13:00, corresponding to solar noon, and shaded areas indicate nighttime periods from sunset to sunrise. The median UHII values are consistently positive across all LCZs. LCZ 8 exhibits the typical daily pattern, with lower UHII values during the day and higher values at night. LCZ 3 shows a similar trend, but the difference between the daytime and nighttime is smaller. In contrast, LCZ 6 experiences higher UHII values during the daytime than at night. In terms of distribution, the UHII exhibits greater variability during the daytime, as indicated by the wider interquartile range. In addition, outliers are much closer to the median at night, reflecting lower nighttime variability. The difference between daytime and nighttime is that air temperature is more influenced by solar radiation during the day, whereas land-cover composition and the more stable atmospheric conditions dominate during the night.

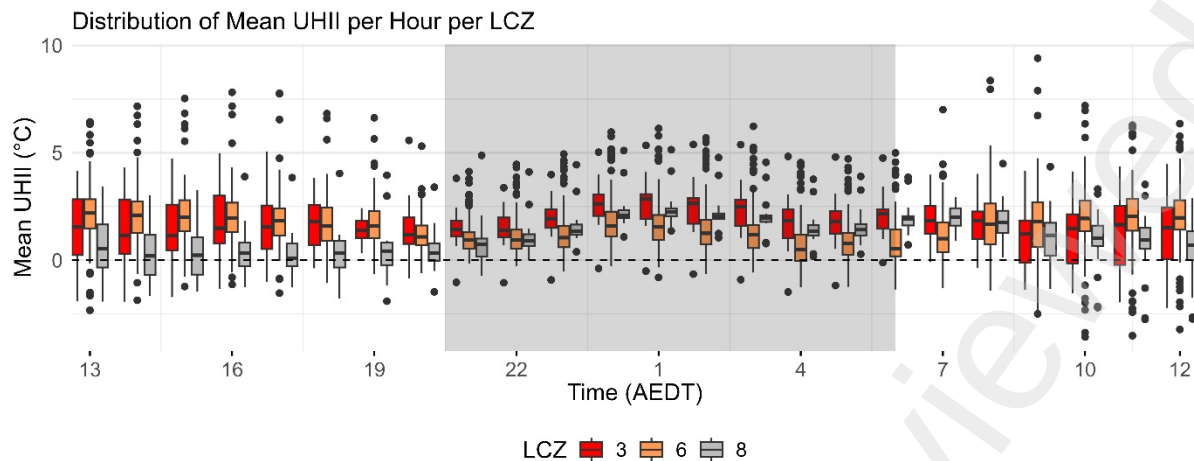


Figure 10: Diurnal distribution of the mean UHII across LCZ 3, 6, and 8 monitoring stations. The x-axis begins at 13:00, corresponding to solar noon. Shaded areas represent nighttime periods from sunset to sunrise.

3.4 Diurnal variation of the UHII and heating rate across days of observation

The diurnal variation of the UHII and the heating/cooling rate, defined as the change in air temperature between consecutive hours, were calculated. Positive heating/cooling rate values indicate heating, while negative values indicate cooling. The heating/cooling rate was compared with that of the reference station. The diurnal variation of the UHII shows how temperature differences between LCZs change throughout the day. **Figure 11** illustrates the diurnal variation and heating/cooling rate of the UHII, beginning at 13:00 AEDT (solar noon). Overall, the UHII is observed clearly at night starting from just after sunset (~20:30 AEDT) to sunrise (~06:00 AEDT). The mean UHII ranged from -2°C to 5°C , with the highest values occurring on 25 and 27 December 2022. These elevated values correspond to consecutive hot days, reflecting the increased heat stress typically associated with heatwave conditions. The UHII was highest at LCZ 3, followed by LCZ 8, LCZ 6, and LCZ D, highlighting the tendency of more compact LCZs to retain heat. Across all LCZs, the cooling/heating rate varies between -2.5°C/h and 3°C/h where positive values occur mostly during the day and negative ones just before sunset.

To evaluate the consistency between the two data sources (PRWS and CWS), the UHII measured at three PRWS stations was used as a reference: LCZ 8 at Melbourne Olympic Park and LCZ D at Essendon Airport and Laverton. Across all dates, UHII values recorded by the CWS were consistently higher than those from the PRWS. This might be caused by the differences in sensor accuracy but also from the greater compactness of settings where CWS are installed compared to the PRWS. As shown in **Supplementary material**, CWS sensors tend to record higher air temperatures (RMSE of around 1°C).

The diurnal pattern of the UHII at the PRWS sites in Essendon Airport and Laverton (LCZ D, the same LCZ type as the rural reference station, Melbourne Airport) reaches a maximum of about 1°C at night. However, because they are surrounded by built-up infrastructure at shorter distances than Melbourne Airport, the UHII is still observed, but with relatively smaller magnitude. Additionally, as shown in the heatmap in **Figure 11**, the UHII recorded at these

PRWS remains slightly above 0°C on most observation days (the air temperature is close to the air temperature recorded at Melbourne Airport). However, noticeable increases are observed during consecutive hot days, specifically on 18-19 and again on 25-27 December 2022 across all LCZs. This indicates that the UHII in LCZ D is generally stable, with increases appearing only during these hot periods, when the rural reference station remains cooler.

The nighttime UHII pattern from PRWS (LCZ 8 at Melbourne Olympic Park) closely resembled that of LCZ 3 and LCZ 8 from the CWS data. Both LCZs 3 and 8 contain more built-up structures than LCZ 6. During the daytime, UHII recorded by the PRWS was lower than that from CWS, likely due to differences in weather station setup. CWS are often located in enclosed areas where limited air circulation restricts heat dissipation, leading to greater heat accumulation compared to PRWS. Daytime patterns for LCZs 3 and 6 were similar; however, LCZ 6 was cooler at night. LCZ 3 has a higher proportion of impervious surfaces, which release stored heat more slowly at night compared to LCZ 6. LCZ 6 heats up rapidly during the day but cools down quickly, often even before sunset.

The heating/cooling rates follow the same pattern for all LCZs and compared to the reference station. The cooling begins around 17:30 AEDT. After sunset, although the heating/cooling rate remains negative, the values begin to rise, indicating a reduction in the cooling rate. Between 22:00 and 01:00 AEDT, the reduction in temperature is more pronounced across all LCZs than at the reference station. During this period, built-up LCZs release the heat accumulated during the day more gradually than the reference station, until around 01:00 AEDT, when all stations, including the rural reference, exhibit similar heating and cooling rates. Following sunrise, air temperatures at the reference station rise more quickly than at all other LCZs. This is likely due to increased shading from buildings in densely built-up LCZs, whereas the reference station experiences greater solar exposure.

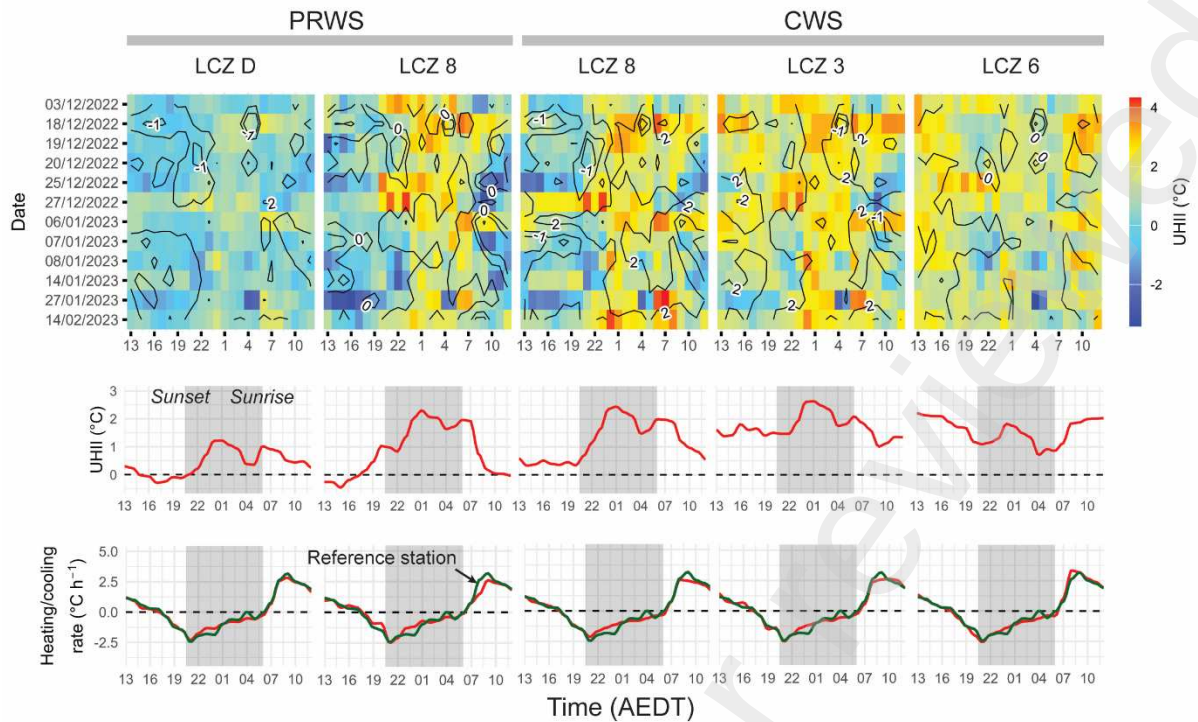


Figure 11: Diurnal cycle of UHII and heating/cooling rates at PRWS and CWS stations

3.5 Model results and validation

Model performance was similar across buffer radii. The daily mean R^2 values ranged from approximately 0.55 to 0.62, with overlapping standard deviations, indicating no clear performance advantage of any single radius. R^2 values were lowest for LCZ 3, followed by LCZ 8 and LCZ 6. For LCZ 3, the 250 m diameter has the highest daily mean R^2 and low RMSE and MAE. The highest R^2 occurs in LCZ 8 and LCZ 6 at the 500 m diameter. **Table 4** presents the daily R^2 , RMSE, and MAE for each LCZ and hexagon size.

Table 4: Daily R^2 , RMSE, and MAE by LCZ and hexagon size

	LCZ 3			LCZ 8			LCZ 6		
Diameter of hexagon (m)	R^2	RMS E	MAE	R^2	RMS E	MAE	R^2	RMS E	MAE
250	0.617	0.208	0.887	0.599	1.190	0.854	0.806	0.881	0.601
500	0.547	0.204	0.941	0.659	1.198	0.858	0.814	0.847	0.580
750	0.594	0.211	0.917	0.615	1.219	0.852	0.806	0.860	0.585
1000	0.609	0.218	0.915	0.587	1.271	0.912	0.801	0.865	0.588

Figure 12 illustrates the hourly variation in R^2 across the LCZ classes. R^2 values for LCZ 3 and LCZ 8 exhibit greater diurnal variability than those for LCZ 6. For LCZ 3, hourly R^2 values indicate that the 250 m hexagon yields the best performance just before sunset and after midnight. Periods starting just before midnight correspond to times when the UHII is most pronounced in LCZ 3, and when land-cover fractions alone cannot explain the increased UHII

without accounting for building height and compactness. For LCZ 8, the 500 m hexagon consistently performs best during both daytime and nighttime. In contrast, for LCZ 6, R^2 curves corresponding to different hexagon diameters largely overlap.

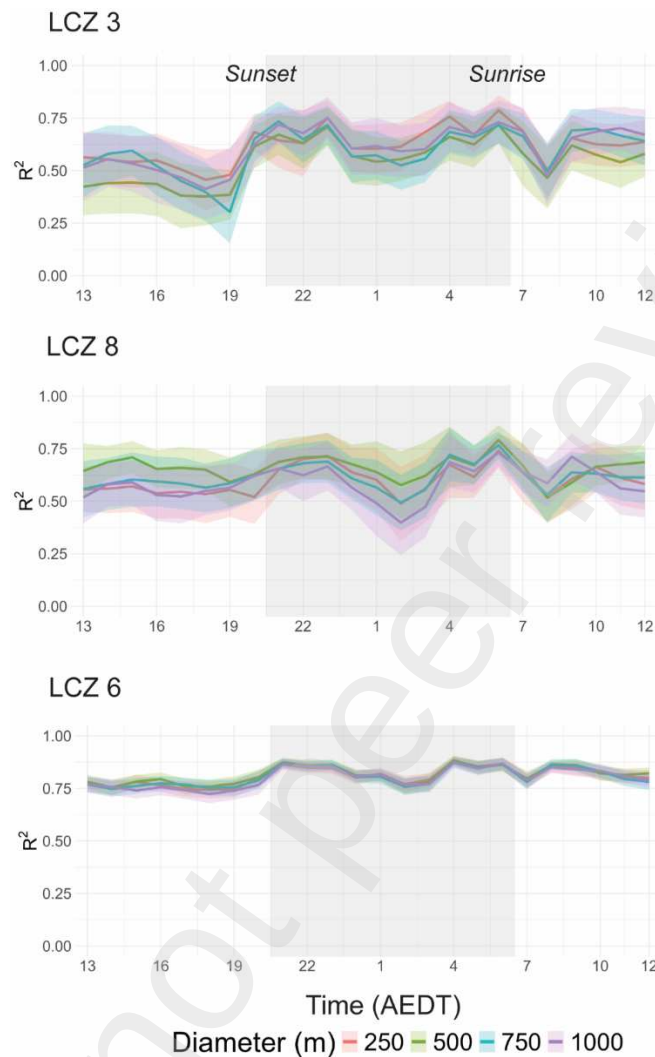


Figure 12: Hourly R^2 for each LCZ

4 Discussion

4.1 Key contributions

This paper focuses on identifying the optimal size of the thermal source area using a data-driven approach with dense weather station observations across three LCZs (3, 6, and 8), by testing hexagonal diameters of 250, 500, 750, and 1000 m. The analysis was conducted at the local scale, in reference to the three main scales used in urban climate studies: microscale, local scale, and mesoscale (Oke, 2025). In Melbourne at this scale, LCZ 6 is the most found LCZ type. Using the RF model, performance metrics (R^2 , RMSE, and MAE) were similar across hexagon diameters. The 500 m hexagon performed slightly better than the others for LCZ 6 and LCZ 8.

For LCZ 3, which features the most dominant built forms and the highest proportion of impervious surfaces, the 250 m hexagon represents the optimal thermal source area size. This indicates that when more compact urban forms are captured within the smaller spatial units, the combined influence of urban form (compact, dense) and surface land-cover characteristics becomes more evident, and the trapped heat can be detected over shorter distances. Stewart et al. (2014) reported that the approximate source-area size in dense urban settings (e.g., city cores) typically encompasses a radius of about 100 m (200 m diameter), whereas in more open areas (e.g., city outskirts), it extends to around 200 m (400 m diameter). In less built-up LCZs, the atmosphere is usually more stable, meaning the air does not mix as much as it does in urban areas. When the air is stable, there is less turbulence (less swirling and mixing of air). Because of this reduced mixing, the source area influencing the measurement at a weather station becomes larger, since the air being measured can come from farther away.

In Lagrangian stochastic dispersion models (Flesch, 1996; Hsieh et al., 1997), the spatial extent of the thermal source area can be quantified using the height-to-fetch ratio. This ratio describes the proportional relationship between the measurement height (z) of an atmospheric sensor and the upwind distance (*fetch*) over which surface conditions significantly influence the observed variable. Under unstable atmospheric conditions, efficient turbulent mixing limits the horizontal transport of air parcels, resulting in relatively small source areas, with typical height-to-fetch ratios of approximately 1:100. In contrast, under neutral and stable conditions, vertical mixing is suppressed, allowing air parcels to travel farther horizontally before reaching the measurement height, leading to larger source areas with reported ratios of approximately 1:250 and 1:300, respectively (Hsieh et al., 2000). A height-to-fetch ratio of 1:100 indicates that the atmospheric signal measured at height z is predominantly influenced by surface conditions extending roughly $100 \times z$ upstream of the sensor.

Multiple data sources were used to assess the UHII, providing a comprehensive view of its spatial and temporal variability. These included 5 PRWS and 132 quality-controlled CWS. Across the study area and its 5 km buffer, this corresponds to an average density of one station every 2.5 km. However, the distribution is uneven, with some locations having very few monitoring sites and others being more densely covered. Previous studies on the use of the LCZ scheme for temperature observations have been conducted in Dublin, Ireland (Alexander & Mills, 2014), Nancy, France (Leconte et al., 2015), Nagpur, India (Kotharkar & Bagade, 2018), and Szeged, Hungary (Skrabit et al., 2017). These studies were limited by the number and spatial/temporal resolution of monitoring stations, and their GIS-based LCZ classifications reduced the generalisability of the findings. By combining a high-density network of weather stations with detailed LCZ classification, this study provides a more robust and locally representative assessment of UHII, highlighting spatial patterns that were previously unresolved.

4.2 Practical implications for urban planning and monitoring

This study confirms that the relationship between UHII and surface land cover fraction is strongest at night, when the effects of stored daytime heat are most pronounced (Oke et al., 2017; Stewart et al., 2014). This reinforces the importance of nighttime monitoring in urban climate assessments and suggests that mitigation measures, such as increasing vegetation or

reflective surfaces, should prioritise reducing nocturnal heat retention. This is particularly relevant in Northern Melbourne suburbs where residents have among the highest heat vulnerability index values in the entire Greater Melbourne metropolitan region (Sun et al., 2019).

Although crowdsourced data are often used for global UHII assessments alongside satellite observations (Demuzere et al., 2022; Fenner et al., 2021), this study demonstrates their underexploited potential for local urban planning. CWS can complement official weather stations by providing dense, fine-scale, and real-time temperature data, enabling cities to pinpoint heat hotspots, prioritise interventions, and track the impact of mitigation measures. The City of Melbourne and its surrounding Local Government Areas have been deploying sensors to detect heat, moisture, temperature, barometric pressure, and other environmental conditions; however, their coverage remains limited. The accessibility of CWS temperature data, combined with LCZ-based representation of urban areas and their surface land cover fractions, can improve preparedness for potential heat hazards and reduce exposure risks.

4.3 Limitations and prospects

4.3.1 Unequal weather station coverage across LCZs

Only three LCZ types were included in this study, and the number of samples among them was unbalanced. Addressing the imbalance was not possible, as most CWS stations are privately owned and located in LCZ 6 and LCZ 3. While not entirely coincidental, the occurrence of a CWS near LCZ 8, which depends on the wind direction, was relatively uncommon. Additionally, LCZ 3 and 8 have a small number of monitoring sites because their extent within the study area was much smaller than that of LCZ 6 after the initial classification from the LCZ Generator. To ensure meaningful model training, the analysis proceeded incrementally, starting with the class with the largest number of samples. This finding highlights the importance of increasing the number of monitoring sites in LCZs with the highest population presence, such as LCZ 3 or other compact and open high-rise LCZs, particularly when planning the placement of public (government-operated) weather sensors in future studies.

4.3.2 Non-uniform measurement heights and variability of surface roughness

Measurements of atmospheric fluxes and scalar quantities such as air temperature are influenced by surface characteristics located upwind of the measurement site. The shape and extent of thermal source areas are primarily controlled by sensor height and by atmospheric turbulence and stability (Voogt & Oke, 2003). In heterogeneous urban environments, surface roughness, driven by obstacles such as buildings and trees, further governs near-surface airflow and the spatial distribution of heat. Measurement height relative to these roughness elements plays a critical role, as higher sensors sample a larger volume of air and therefore integrate signals from broader source areas (Schmid, 1994).

Although a standardized measurement height would be ideal for ensuring comparability, this is only feasible using mobile weather stations over limited spatial extents and is constrained by temporal temperature variability during data collection. At the local scale, professional weather

stations offer high accuracy but are costly and difficult to deploy in dense urban settings. As a result, dense networks of citizen weather stations provide a practical alternative for capturing urban temperature variability, but their heterogeneous measurement heights introduce uncertainty in the size and representativeness of the associated thermal source areas.

The process of deciding how and where to install a weather station is called "Siting". As a reference station, the anemometer at Melbourne Airport is well-sited, located next to the runways, with over 400 m clearance from the main runway, terminals, and other paved surfaces (LCZ 8 and LCZ E, bare rock or paved), with wind measurements taken at the standard height of 10 m (Holmes, 2022). On the other hand, the metadata of CWS stations provide only the ground elevation above sea level, without information on the sensor's mounting height. According to installation guidelines, sensors are typically placed at least 1.5 m above the surface, whether on soil or rooftops.

Wunderground's installation guide provides theoretical advice, recommending sensors be placed at least 25 m away from tall objects such as trees and buildings. In contrast, Netatmo reflects more realistic conditions, offering flexible setup options such as mounting on a pole or wall. However, Nipen et al. (2020) explain that most sources of error, such as direct sunlight or proximity to walls, tend to produce a warm bias in the measurements from CWS, which was also observed in our data. Meteorological standards specify that air temperature should be measured between 1 and 1.5 m above the ground in a shaded, well-ventilated environment, avoiding placement above concrete or dark surfaces (Overton, 2007). These ideal conditions are rarely met in urban settings, where residents often have trees, buildings, and other obstructions near their measurement sites.

Consequently, it is acknowledged that the siting and setup of weather stations, especially CWS, which were predominantly used in this research, play a decisive role in the reliability of the results. While the inherent heterogeneity of urban environments introduces local variability in temperature measurements, it is not necessarily problematic. Such variability reflects the distinct microclimatic conditions that each station captures within its own source area. Recognizing these differences is essential for interpreting spatial temperature patterns and for understanding how local surface characteristics influence observed thermal conditions.

4.3.3 Topography and the applicability of LCZs

This study has described how the components of an urban area interact to shape the local climate using a data-driven approach. Extending this principle to larger regions would allow these interactions to be analysed at the city scale, considering the combined effects of different LCZs. However, one parameter not addressed in this study is topography, although the study area consists mainly of gently sloping terrain. Topography can significantly influence near-surface climate by affecting the exchange of energy, water, and momentum between the land surface and atmosphere (Raupach & Finnigan, 1997). It is important to note, however, that the primary purpose of LCZs is not for detailed climate analysis, which requires local knowledge of topography, but rather to generalise the characteristics of built forms and land cover types that are universally recognised, rather than reflecting city-specific climatology (Stewart & Oke, 2012). In addition, feature importance analysis by Demuzere et al. (2019) for European cities

showed that aspect, slope, and terrain had low importance (below 0.4) in LCZ classification. In summary, while LCZs provide a useful framework for generalizing urban form and land cover characteristics, their limitations - including the lack of explicit topographic detail - should be acknowledged. For relatively flat study areas like the one analysed here, these limitations are less critical, and LCZ-based analyses can still effectively capture the influence of urban structure on local climate patterns.

5 Conclusion

From Tobler's First Law of Geography to the determination of the optimal scale of the thermal source area, this study quantified the influence of urban form and surface land cover on UHII across multiple spatial and temporal scales, aiming to determine how the estimated size of thermal source areas varies throughout the day and by LCZ type. The incorporation of urban climate knowledge into city planning has achieved limited effectiveness, in part because urban climatology has progressed slowly with respect to issues of scale (Stewart & Oke, 2012). The approach of defining the optimum size of the thermal source area is well-grounded in physical reasoning, rather than relying on arbitrary units such as city blocks or fixed pixel sizes.

The integration of diverse data sources, such as LCZs derived from the global LCZ Generator produced by WUDAPT, quality-controlled air temperature measurements from CWS and PRWS, and high-spatial-resolution land cover data from the CSIRO Urban Monitor, enabled a data-driven approach to analysing UHII. Across LCZs with higher building density, characterised by taller structures, greater compactness, and extensive use of concrete, such as LCZs 3 and 8, UHII is most pronounced during the nighttime. The Random Forest model indicated that a 500 m-diameter hexagon best captures the combined thermal influence of open (LCZ 6) and large low-rise (LCZ 8) areas, while a 250 m hexagon is optimal for compact low-rise (LCZ 3). These findings are consistent with the physically derived definition of the thermal source area, effectively bridging LCZ definitions derived from remote sensing techniques with physics-based approaches grounded in surface characteristics.

Capturing these spatial and temporal dynamics of the UHII requires high-resolution data, which is often limited when relying solely on traditional official weather stations. The use of CWS offers both advantages and challenges: although their measurements are quality-controlled, accuracy may be affected by non-standard siting compared to PRWS; however, this also allows them to better represent real-world conditions in urban environments. Overall, the study demonstrated that integrating CWS, PRWS, land cover analysis, and machine learning provided a practical, data-driven framework for understanding the thermal source area. The use of an extended network of Citizen Weather Stations could potentially increase sample size and reliability, while also aiding in the identification and practical implementation of interventions, such as tree planting or surface modifications, to maximise cooling benefits across the city.

Data availability

Data will be available upon request. Some citizen weather data are subject to a signed agreement with the data provider and therefore cannot be publicly shared.

References

- Acosta, M. P., Vahdatikhaki, F., Santos, J., Hammad, A., & Doree, A. G. (2021). How to bring UHI to the urban planning table? A data-driven modeling approach. *Sustainable Cities and Society*, 71, 102948. <https://doi.org/10.1016/j.scs.2021.102948>
- Acosta, M. P., Vahdatikhaki, F., Santos, J., Jarro, S. P., & Dorée, A. G. (2024). Data-driven analysis of Urban Heat Island phenomenon based on street typology. *Sustainable Cities and Society*, 101, 105170. <https://doi.org/10.1016/j.scs.2023.105170>
- Adilkhanova, I., Ngarambe, J., & Yun, G. Y. (2022). Recent advances in black box and white-box models for urban heat island prediction: Implications of fusing the two methods. *Renewable and Sustainable Energy Reviews*, 165, 112520. <https://doi.org/10.1016/j.rser.2022.112520>
- Al-Obaidi, I., Rayburg, S., Pórolniczak, M., & Neave, M. (2021). Assessing the impact of wind conditions on urban heat islands in large Australian cities. *Journal of Ecological Engineering*, 22(11), 1-15. <https://doi.org/10.12911/22998993/142967>
- Alexander, P. J., & Mills, G. (2014). Local climate classification and Dublin's urban heat island. *Atmosphere*, 5(4), 755-774. <https://doi.org/10.3390/atmos5040755>
- Arnfield, A. J. (2003). Two decades of urban climate research: a review of turbulence, exchanges of energy and water, and the urban heat island. *International Journal of Climatology: A Journal of the Royal Meteorological Society*, 23(1), 1-26. <https://doi.org/10.1002/joc.859>
- Bechtel, B., Alexander, P. J., Beck, C., Böhner, J., Brousse, O., Ching, J., Demuzere, M., Fonte, C., Gál, T., Hidalgo, J., Hoffmann, P., Middel, A., Mills, G., Ren, C., See, L., Sismanidis, P., Verdonck, M.-L., Xu, G., & Xu, Y. (2019). Generating WUDAPT Level 0 data – Current status of production and evaluation. *Urban Climate*, 27, 24-45. <https://doi.org/https://doi.org/10.1016/j.uclim.2018.10.001>
- Birch, C. P. D., Oom, S. P., & Beecham, J. A. (2007). Rectangular and hexagonal grids used for observation, experiment and simulation in ecology. *Ecological Modelling*, 206(3), 347-359. <https://doi.org/https://doi.org/10.1016/j.ecolmodel.2007.03.041>
- Breiman, L. (2001). Random forests. *Machine learning*, 45(1), 5-32.
- Carr, D. B., Olsen, A. R., & White, D. (1992). Hexagon mosaic maps for display of univariate and bivariate geographical data. *Cartography and Geographic Information Systems*, 19(4), 228-236. <https://doi.org/10.1559/152304092783721231>
- Chen, Y., & Hu, Y. (2022). The urban morphology classification under local climate zone scheme based on the improved method - A case study of Changsha, China. *Urban Climate*, 45, 101271. <https://doi.org/https://doi.org/10.1016/j.uclim.2022.101271>
- Ching, J., Mills, G., Bechtel, B., See, L., Feddema, J., Wang, X., Ren, C., Brousse, O., Martilli, A., & Neophytou, M. (2018). WUDAPT: An urban weather, climate, and environmental modeling infrastructure for the anthropocene. *Bulletin of the American Meteorological Society*, 99(9), 1907-1924. <https://doi.org/10.1175/BAMS-D-16-0236.1>
- Cutler, D. R., Edwards Jr, T. C., Beard, K. H., Cutler, A., Hess, K. T., Gibson, J., & Lawler, J. J. (2007). Random forests for classification in ecology. *Ecology*, 88(11), 2783-2792. <https://doi.org/10.1890/07-0539.1>
- Demuzere, M., Bechtel, B., Middel, A., & Mills, G. (2019). Mapping Europe into local climate zones. *PloS one*, 14(4), e0214474. <https://doi.org/10.1371/journal.pone.0214474>
- Demuzere, M., Kittner, J., & Bechtel, B. (2021). LCZ Generator: a web application to create Local Climate Zone maps. *Frontiers in Environmental Science*, 9, 637455. <https://doi.org/10.3389/fenvs.2021.637455>
- Demuzere, M., Kittner, J., Martilli, A., Mills, G., Moede, C., Stewart, I. D., van Vliet, J., & Bechtel, B. (2022). A global map of Local Climate Zones to support earth system

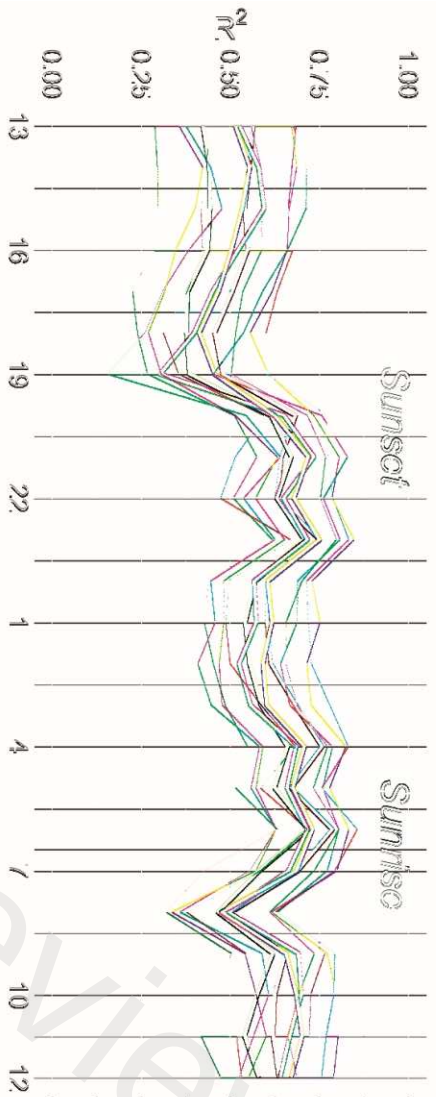
- modelling and urban scale environmental science. *Earth System Science Data Discussions*, 2022, 1-57. <https://doi.org/10.5194/essd-14-3835-2022>
- Dimitrov, S., Popov, A., & Iliev, M. (2021). An Application of the LCZ Approach in Surface Urban Heat Island Mapping in Sofia, Bulgaria. *Atmosphere*, 12(11), 1370. <https://www.mdpi.com/2073-4433/12/11/1370>
- Erell, E., & Williamson, T. (2007). Intra-urban differences in canopy layer air temperature at a mid-latitude city. *International Journal of Climatology: A Journal of the Royal Meteorological Society*, 27(9), 1243-1255. <https://doi.org/10.1002/joc.1469>
- Fenner, D., Bechtel, B., Demuzere, M., Kittner, J., & Meier, F. (2021). CrowdQC+—a quality-control for crowdsourced air-temperature observations enabling world-wide urban climate applications. *Frontiers in Environmental Science*, 9, 720747. <https://doi.org/10.3389/fenvs.2021.720747>
- Flesch, T. K. (1996). The footprint for flux measurements, from backward Lagrangian stochastic models. *Boundary-Layer Meteorology*, 78(3), 399-404. <https://doi.org/10.1007/BF00120943>
- Grimmond, C. S. B., Blackett, M., Best, M. J., Barlow, J., Baik, J., Belcher, S., Bohnenstengel, S., Calmet, I., Chen, F., & Dandou, A. (2010). The international urban energy balance models comparison project: first results from phase 1. *Journal of Applied Meteorology and Climatology*, 49(6), 1268-1292. <https://doi.org/10.1175/2010JAMC2354.1>
- Holmes, J. (2022). Wind climate of the Melbourne metropolitan area. *Proceedings of the Royal Society of Victoria*, 133(2), 82-92. <https://doi.org/10.1071/RS21011>
- Howard, L. (1833). *The climate of London: deduced from meteorological observations made in the metropolis and at various places around it* (Vol. 3). Harvey and Darton, J. and A. Arch, Longman, Hatchard, S. Highley [and] R. Hunter.
- Hsieh, C.-I., Katul, G., & Chi, T.-w. (2000). An approximate analytical model for footprint estimation of scalar fluxes in thermally stratified atmospheric flows. *Advances in Water Resources*, 23(7), 765-772. [https://doi.org/10.1016/S0309-1708\(99\)00042-1](https://doi.org/10.1016/S0309-1708(99)00042-1)
- Hsieh, C. I., Katul, G. G., Schieldge, J., Sigmon, J. T., & Knoerr, K. K. (1997). The Lagrangian stochastic model for fetch and latent heat flux estimation above uniform and nonuniform terrain. *Water Resources Research*, 33(3), 427-438. <https://doi.org/10.1029/96WR03136>
- Hurley, J., Saunders, M. A., Both, A., Sun, C., Boruff, B., Duncan, J., Amati, M., & Caccetta, P. (2019). Urban vegetation cover change in Melbourne. *Centre for Urban Research, RMIT University: Melbourne, Australia*.
- Keay, K., & Simmonds, I. (2005). The association of rainfall and other weather variables with road traffic volume in Melbourne, Australia. *Accident analysis & prevention*, 37(1), 109-124. <https://doi.org/10.1016/j.aap.2004.07.005>
- Kim, Y.-H., & Baik, J.-J. (2002). Maximum urban heat island intensity in Seoul. *Journal of applied meteorology*, 41(6), 651-659. [https://doi.org/10.1175/1520-0450\(2002\)041<0651:MUHIII>2.0.CO;2](https://doi.org/10.1175/1520-0450(2002)041<0651:MUHIII>2.0.CO;2)
- Kljun, N., Rotach, M., & Schmid, H. (2002). A three-dimensional backward Lagrangian footprint model for a wide range of boundary-layer stratifications. *Boundary-Layer Meteorology*, 103, 205-226. <https://doi.org/10.1023/A:1014556300021>
- Kotharkar, R., & Bagade, A. (2018). Evaluating urban heat island in the critical local climate zones of an Indian city. *Landscape and Urban Planning*, 169, 92-104. <https://doi.org/10.1016/j.landurbplan.2017.08.009>
- Kottek, M., Grieser, J., Beck, C., Rudolf, B., & Rubel, F. (2006). World map of the Köppen-Geiger climate classification updated. <https://doi.org/10.1127/0941-2948/2006/0130>

- Leconte, F., Bouyer, J., Claverie, R., & Pétrissans, M. (2015). Using Local Climate Zone scheme for UHI assessment: Evaluation of the method using mobile measurements. *Building and environment*, 83, 39-49. <https://doi.org/10.1016/j.buildenv.2014.05.005>
- Li, D., Sun, T., Liu, M., Yang, L., Wang, L., & Gao, Z. (2015). Contrasting responses of urban and rural surface energy budgets to heat waves explain synergies between urban heat islands and heat waves. *Environmental Research Letters*, 10(5), 054009. <https://doi.org/10.1088/1748-9326/10/5/054009>
- Liu, L., Lin, Y., Liu, J., Wang, L., Wang, D., Shui, T., Chen, X., & Wu, Q. (2017). Analysis of local-scale urban heat island characteristics using an integrated method of mobile measurement and GIS-based spatial interpolation. *Building and environment*, 117, 191-207. <https://doi.org/10.1016/j.buildenv.2017.03.013>
- Liu, S., & Shi, Q. (2020). Local climate zone mapping as remote sensing scene classification using deep learning: A case study of metropolitan China. *ISPRS Journal of Photogrammetry and Remote Sensing*, 164, 229-242. <https://doi.org/10.1016/j.isprsjprs.2020.04.008>
- Lizana, J., Núñez-Peiró, M., Khosla, R., & McCulloch, M. (2023). Citizen data for global mapping of atmospheric urban heat islands. <https://doi.org/10.21203/rs.3.rs-2924752/v1>
- Mirzaei, P. A. (2015). Recent challenges in modeling of urban heat island. *Sustainable Cities and Society*, 19, 200-206. <https://doi.org/10.1016/j.scs.2015.04.001>
- Montávez, J. P., Rodríguez, A., & Jiménez, J. I. (2000). A study of the urban heat island of Granada. *International Journal of Climatology: A Journal of the Royal Meteorological Society*, 20(8), 899-911. [https://doi.org/10.1002/1097-0088\(20000630\)20:8<899::AID-JOC433>3.0.CO;2-I](https://doi.org/10.1002/1097-0088(20000630)20:8<899::AID-JOC433>3.0.CO;2-I)
- Morris, C., Simmonds, I., & Plummer, N. (2001). Quantification of the influences of wind and cloud on the nocturnal urban heat island of a large city. *Journal of applied meteorology*, 40(2), 169-182. [https://doi.org/10.1175/1520-0450\(2001\)040<0169:QOTIOW>2.0.CO;2](https://doi.org/10.1175/1520-0450(2001)040<0169:QOTIOW>2.0.CO;2)
- Nipen, T. N., Seierstad, I. A., Lussana, C., Kristiansen, J., & Hov, Ø. (2020). Adopting citizen observations in operational weather prediction. *Bulletin of the American Meteorological Society*, 101(1), E43-E57. <https://doi.org/10.1175/BAMS-D-18-0237.1>
- Noi, P. T., Degener, J., & Kappas, M. (2017). Comparison of multiple linear regression, cubist regression, and random forest algorithms to estimate daily air surface temperature from dynamic combinations of MODIS LST data. *Remote Sensing*, 9(5), 398. <https://doi.org/10.3390/rs9050398>
- Núñez Peiró, M., Sánchez-Guevara Sánchez, C., & Neila González, F. J. (2019). Source area definition for local climate zones studies. A systematic review. *Building and environment*, 148, 258-285. <https://doi.org/10.1016/j.buildenv.2018.10.050>
- Oke, T. R. (1973). City size and the urban heat island. *Atmospheric Environment* (1967), 7(8), 769-779. [https://doi.org/10.1016/0004-6981\(73\)90140-6](https://doi.org/10.1016/0004-6981(73)90140-6)
- Oke, T. R. (1982). The energetic basis of the urban heat island. *Quarterly journal of the royal meteorological society*, 108(455), 1-24.
- Oke, T. R. (1987). Boundary Layer Climates. In (pp. 435): Routledge.
- Oke, T. R. (1988). The urban energy balance. *Progress in Physical Geography*, 12(4), 471-508. <https://doi.org/10.1177/030913338801200401>
- Oke, T. R. (2025). Initial guidance to obtain representative meteorological observations at urban sites.
- Oke, T. R., Mills, G., Christen, A., & Voogt, J. A. (2017). *Urban climates*. Cambridge university press.

- Oukawa, G. Y., Krecl, P., & Targino, A. C. (2022). Fine-scale modeling of the urban heat island: A comparison of multiple linear regression and random forest approaches. *Science of the Total Environment*, 815, 152836. <https://doi.org/https://doi.org/10.1016/j.scitotenv.2021.152836>
- Overton, A. K. (2007). A guide to the siting, exposure and calibration of automatic weather stations for synoptic and climatological observations. In: Royal Meteorological Society Reading.
- Raupach, M., & Finnigan, J. (1997). The influence of topography on meteorological variables and surface-atmosphere interactions. *Journal of Hydrology*, 190(3-4), 182-213. [https://doi.org/10.1016/S0022-1694\(96\)03127-7](https://doi.org/10.1016/S0022-1694(96)03127-7)
- Rogers, C. D. W., Gallant, A. J. E., & Tapper, N. J. (2019). Is the urban heat island exacerbated during heatwaves in southern Australian cities? *Theoretical and applied climatology*, 137(1), 441-457. <https://doi.org/10.1007/s00704-018-2599-x>
- Romero Rodríguez, L., Sánchez Ramos, J., & Álvarez Domínguez, S. (2023). Simplifying the process to perform air temperature and UHI measurements at large scales: Design of a new APP and low-cost Arduino device. *Sustainable Cities and Society*, 95, 104614. <https://doi.org/https://doi.org/10.1016/j.scs.2023.104614>
- Runnalls, K., & Oke, T. (2000). Dynamics and controls of the near-surface heat island of Vancouver, British Columbia. *Physical Geography*, 21(4), 283-304. <https://doi.org/10.1080/02723646.2000.10642711>
- Sachindra, D., Ng, A., Muthukumaran, S., & Perera, B. (2016). Impact of climate change on urban heat island effect and extreme temperatures: a case-study. *Quarterly journal of the royal meteorological society*, 142(694), 172-186. <https://doi.org/10.1002/qj.2642>
- Schmid, H. (1994). Source areas for scalars and scalar fluxes. *Boundary-Layer Meteorology*, 67(3), 293-318. <https://doi.org/10.1007/BF00713146>
- Schmid, H. P. (2002). Footprint modeling for vegetation atmosphere exchange studies: a review and perspective. *Agricultural and forest meteorology*, 113(1-4), 159-183. [https://doi.org/10.1016/S0168-1923\(02\)00107-7](https://doi.org/10.1016/S0168-1923(02)00107-7)
- Shepard, D. (1968). A two-dimensional interpolation function for irregularly-spaced data. Proceedings of the 1968 23rd ACM national conference. <https://doi.org/10.1145/800186.810616>
- Skrabit, N., Stewart, I. D., Unger, J., & Gál, T. M. (2017). Employing an urban meteorological network to monitor air temperature conditions in the 'local climate zones' of Szeged, Hungary. *International journal of climatology*, 37(S1), 582-596. <https://doi.org/10.1002/joc.5023>
- Stewart, I. D. (2011). *Redefining the urban heat island* University of British Columbia.
- Stewart, I. D., & Oke, T. R. (2012). Local climate zones for urban temperature studies. *Bulletin of the American Meteorological Society*, 93(12), 1879-1900. <https://doi.org/10.1175/BAMS-D-11-00019.1>
- Stewart, I. D., Oke, T. R., & Kravynhoff, E. S. (2014). Evaluation of the 'local climate zone' scheme using temperature observations and model simulations. *International journal of climatology*, 34(4). <https://doi.org/10.1002/joc.3746>
- Sundborg, Å. (1951). Climatological studies in Uppsala: with special regard to the temperature conditions in the urban area. (No Title).
- Tobler, W. R. (1970). A computer movie simulating urban growth in the Detroit region. *Economic geography*, 46(sup1), 234-240. <https://doi.org/10.2307/143141>
- Vavassori, A., Giuliani, G., & Brovelli, M. A. (2023). Mapping Local Climate Zones in Lausanne (Switzerland) with Sentinel-2 and PRISMA imagery: comparison of classification performance using different band combinations and building height data.

- International Journal Of Digital Earth*, 16(2), 4790-4810.
<https://doi.org/10.1080/17538947.2023.2283485>
- Voelkel, J., & Shandas, V. (2017). Towards systematic prediction of urban heat islands: Grounding measurements, assessing modeling techniques. *Climate*, 5(2), 41.
<https://doi.org/10.3390/cli5020041>
- Voogt, J. A., & Oke, T. R. (2003). Thermal remote sensing of urban climates. *Remote Sensing of Environment*, 86(3), 370-384. [https://doi.org/10.1016/S0034-4257\(03\)00079-8](https://doi.org/10.1016/S0034-4257(03)00079-8)
- Wellinger, N., Gubler, M., Müller, F., & Brönnimann, S. (2024). GIS-based revision of a WUDAPT Local Climate Zones map of Bern, Switzerland. *City and Environment Interactions*, 21, 100135. <https://doi.org/https://doi.org/10.1016/j.cacint.2023.100135>
- Wieringa, J. (1993). Representative roughness parameters for homogeneous terrain. *Bound.-Layer Meteor*, 63, 323-363. <https://doi.org/10.1007/BF00705357>

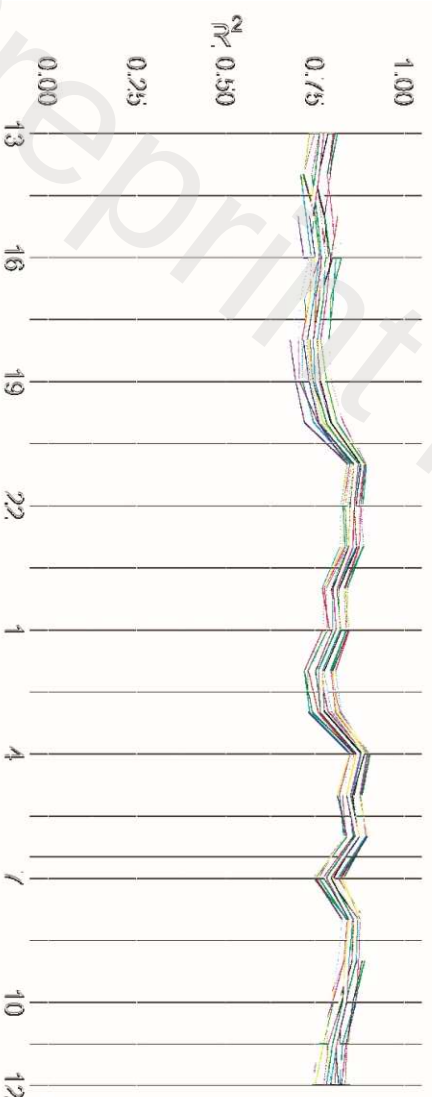
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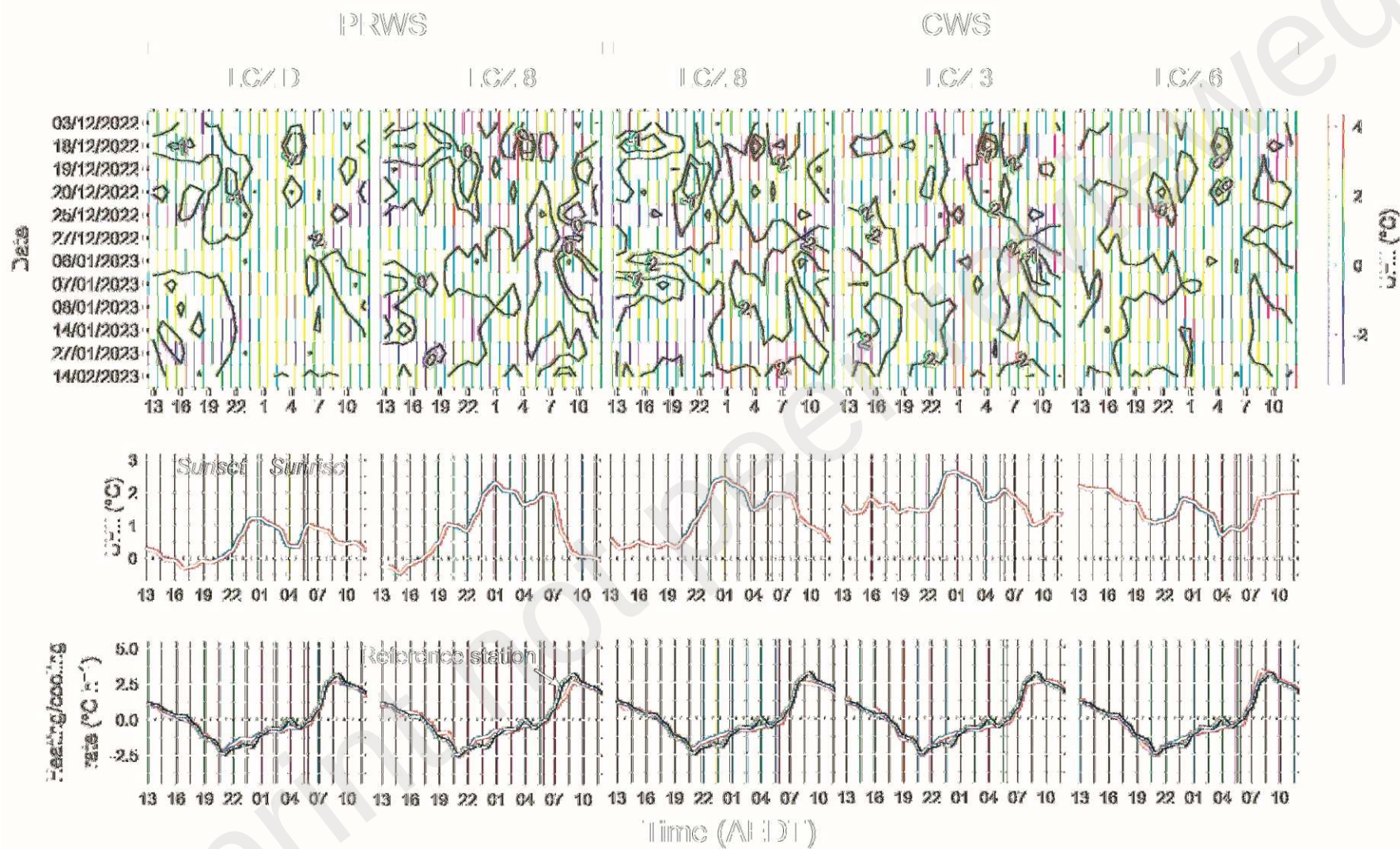
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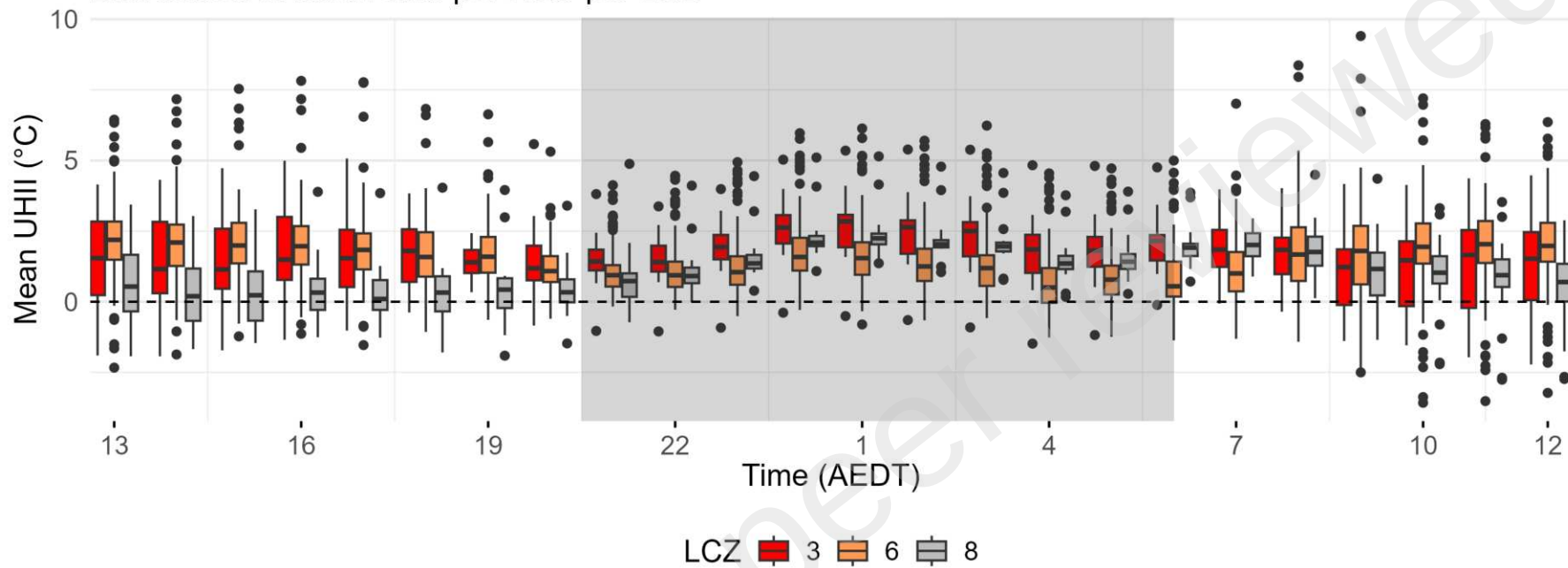
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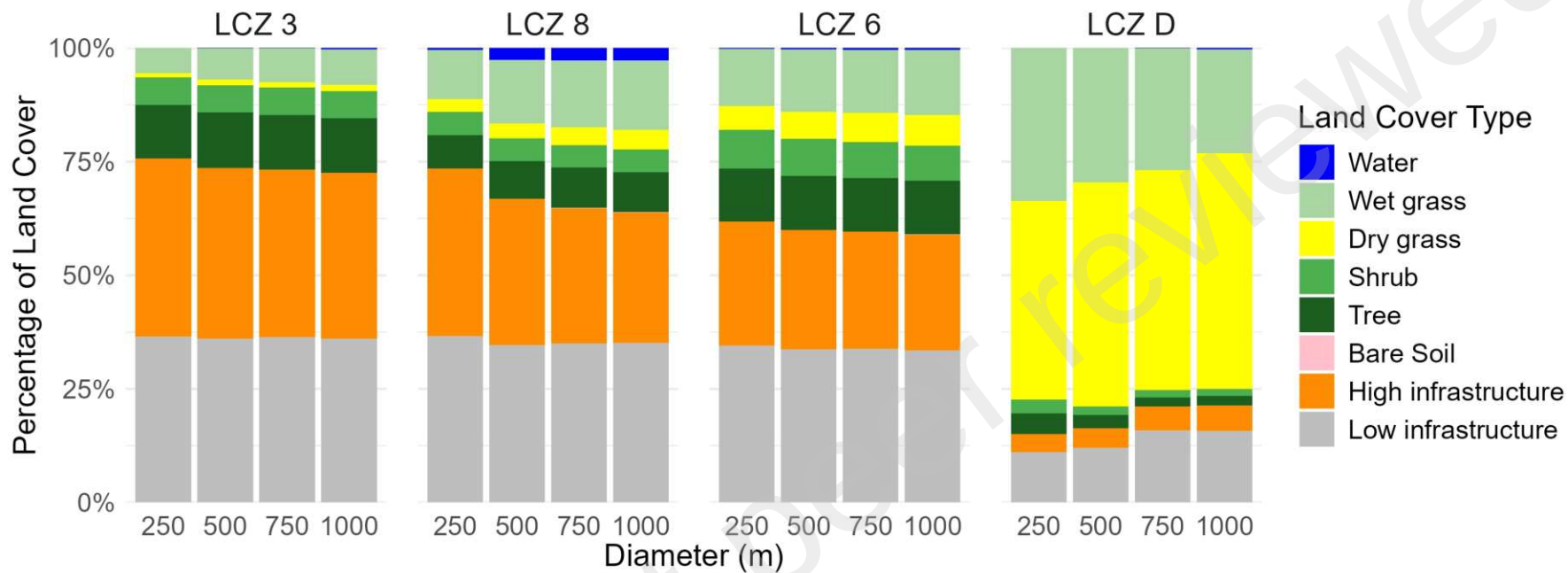
Diameter (m) | 250 | 500 | 750 | 1000

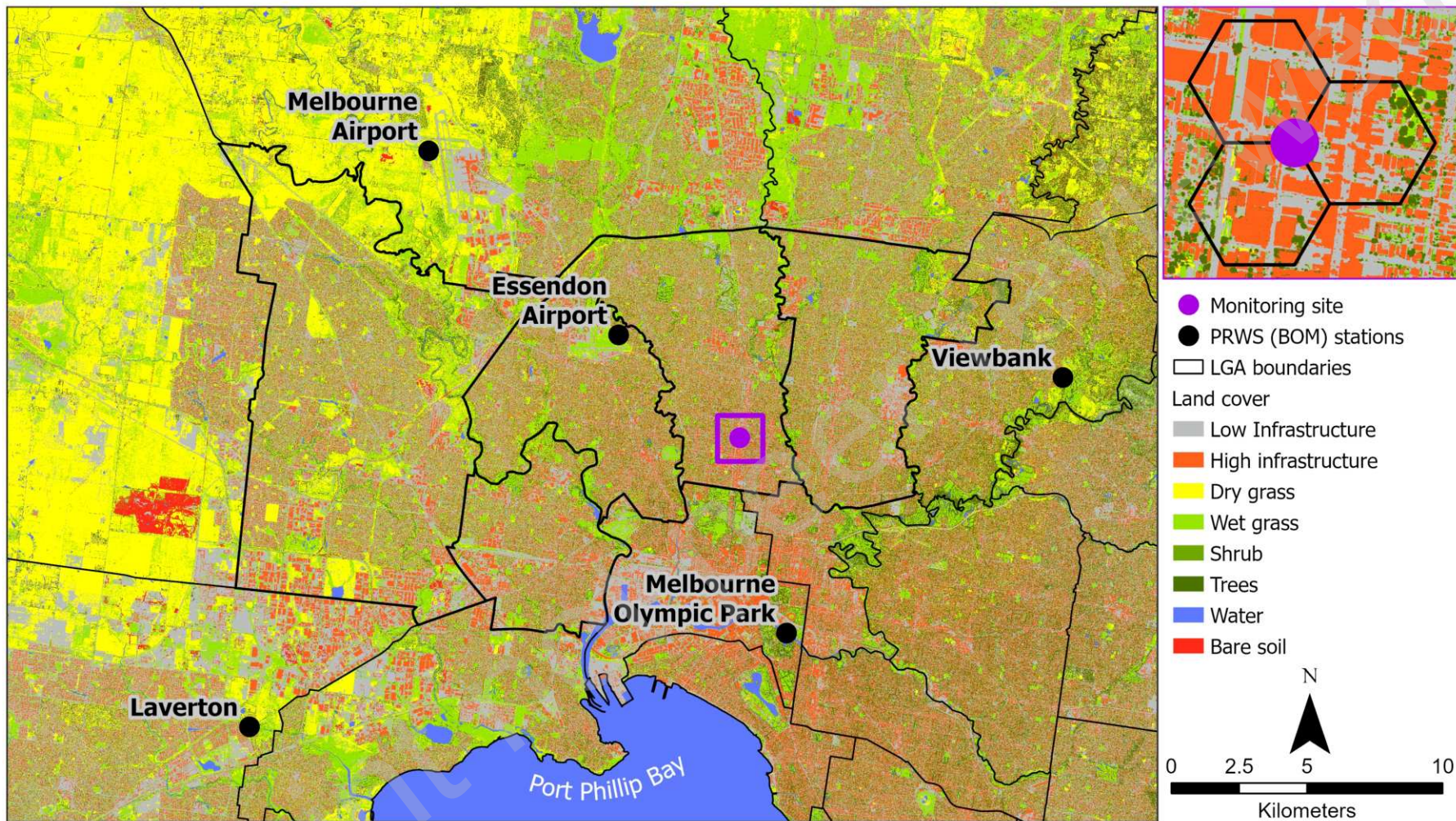


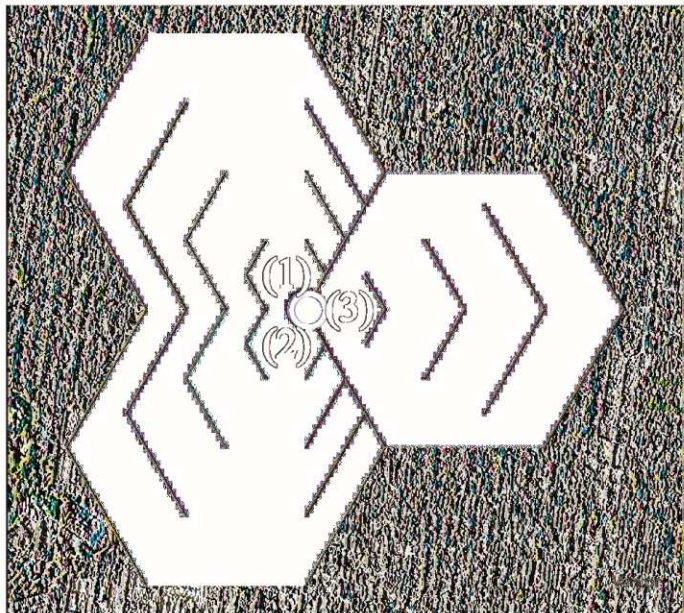
Distribution of Mean UHII per Hour per LCZ



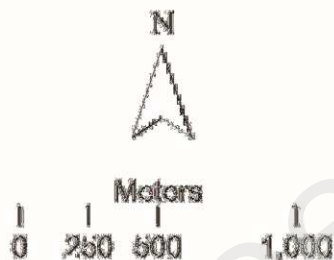
Surface cover fraction by Diameter and LCZ



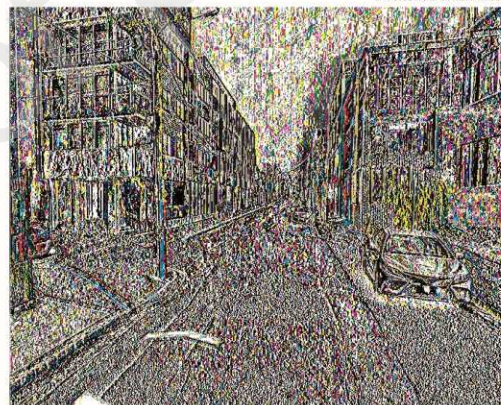
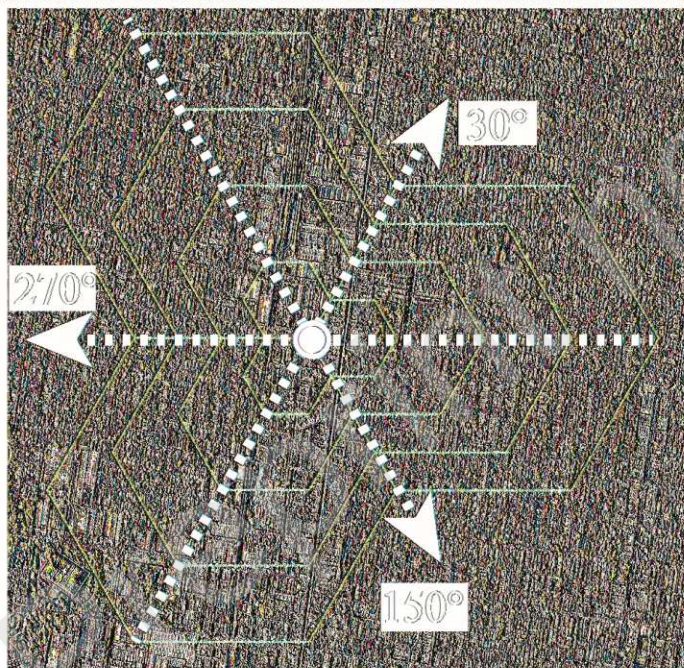
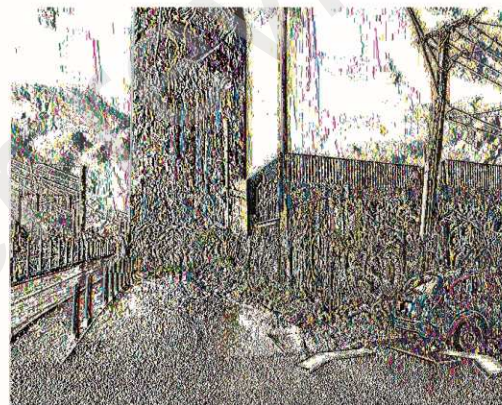




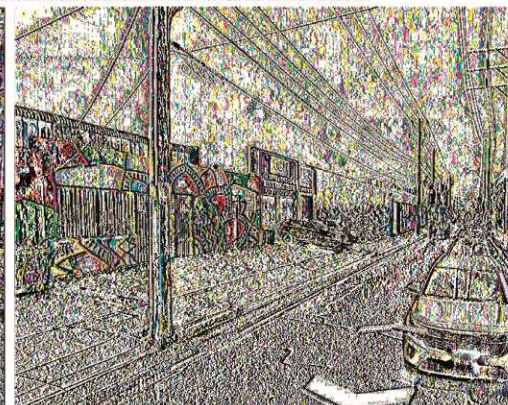
- Monitoring site
- LC
- 13. Compact lowrise
- 16. Open lowrise
- 18. Large lowrise



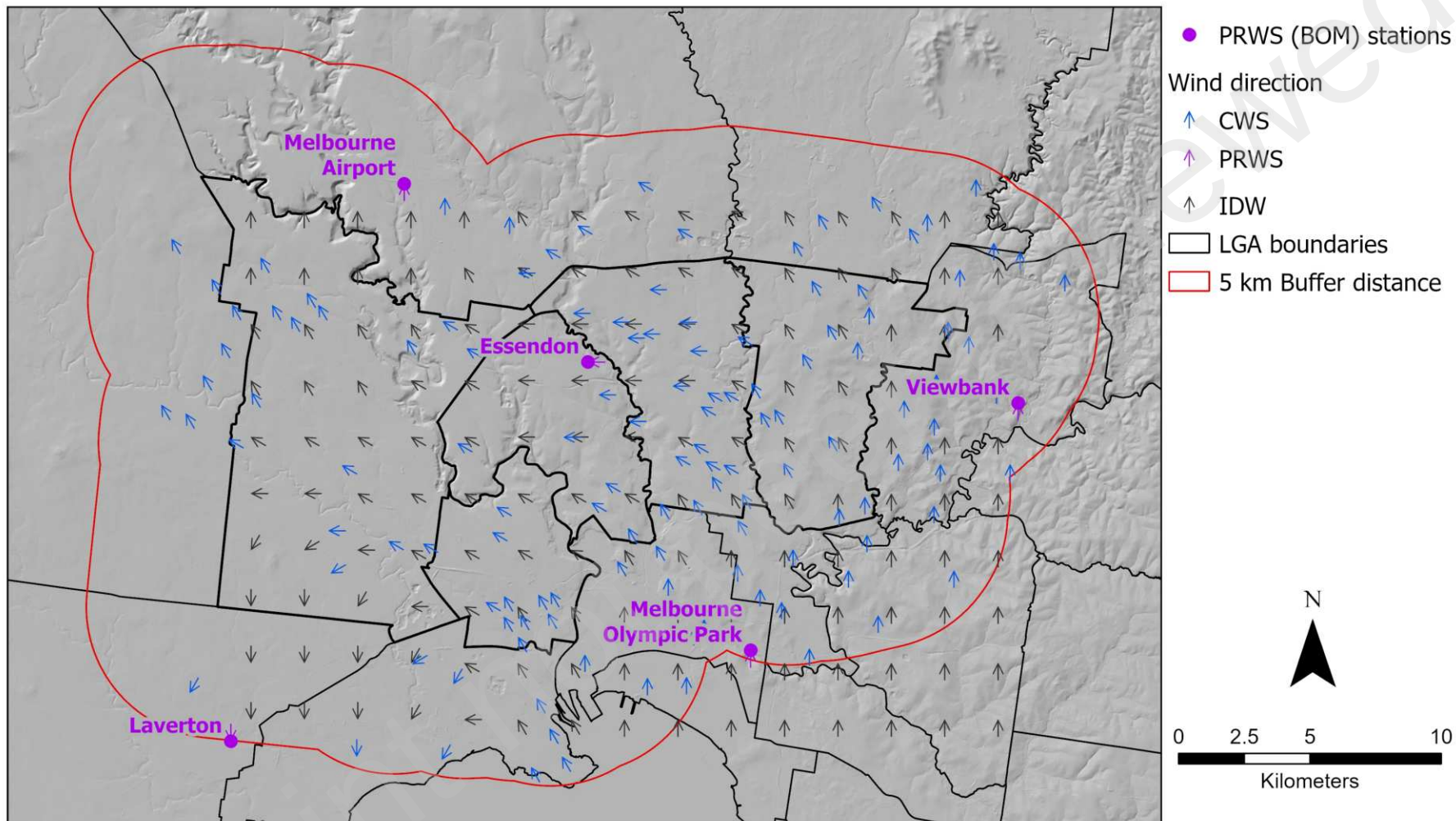
(1): LC/ 8. Large lowrise

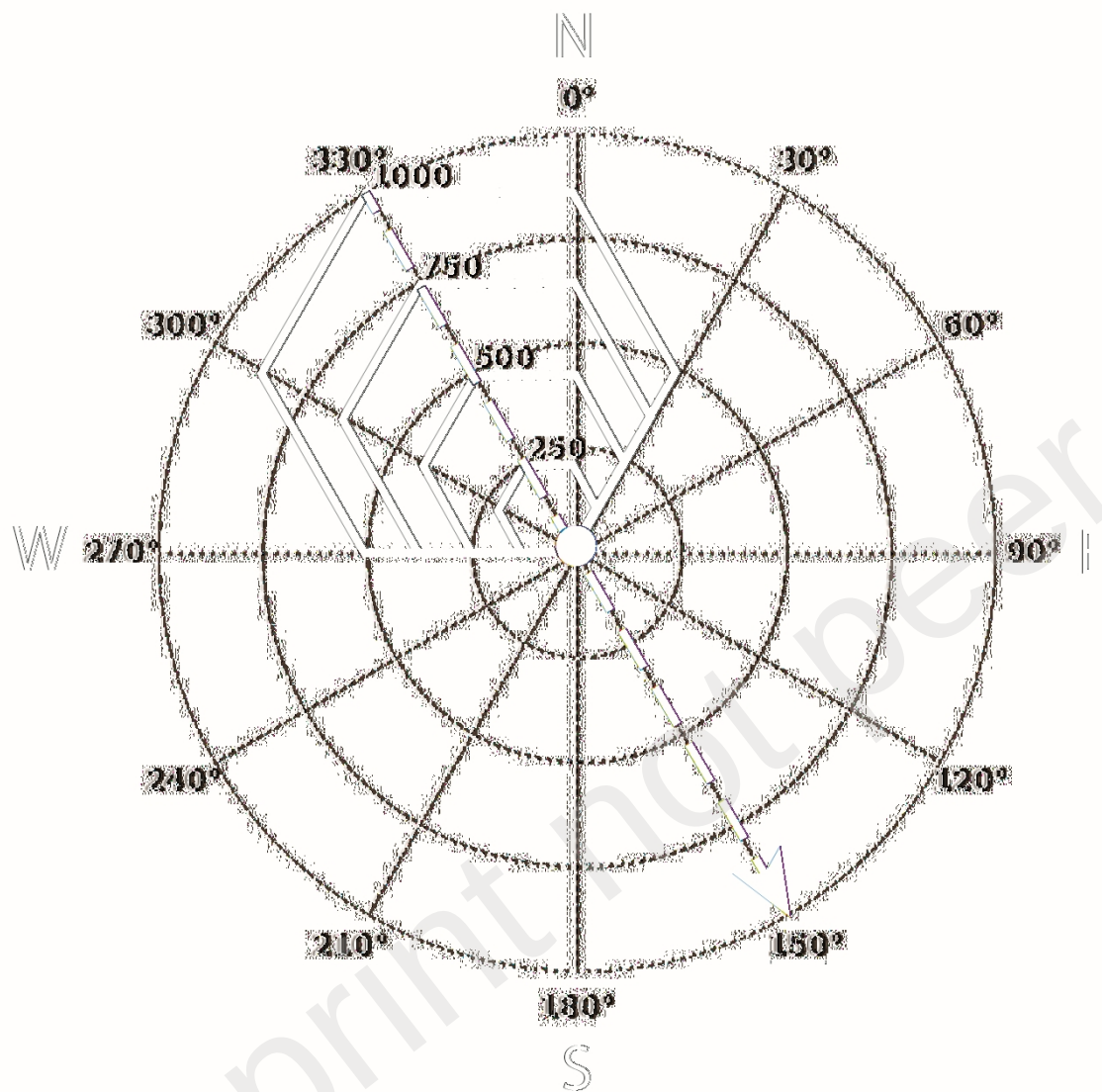


(2): LC/ 3. Compact lowrise

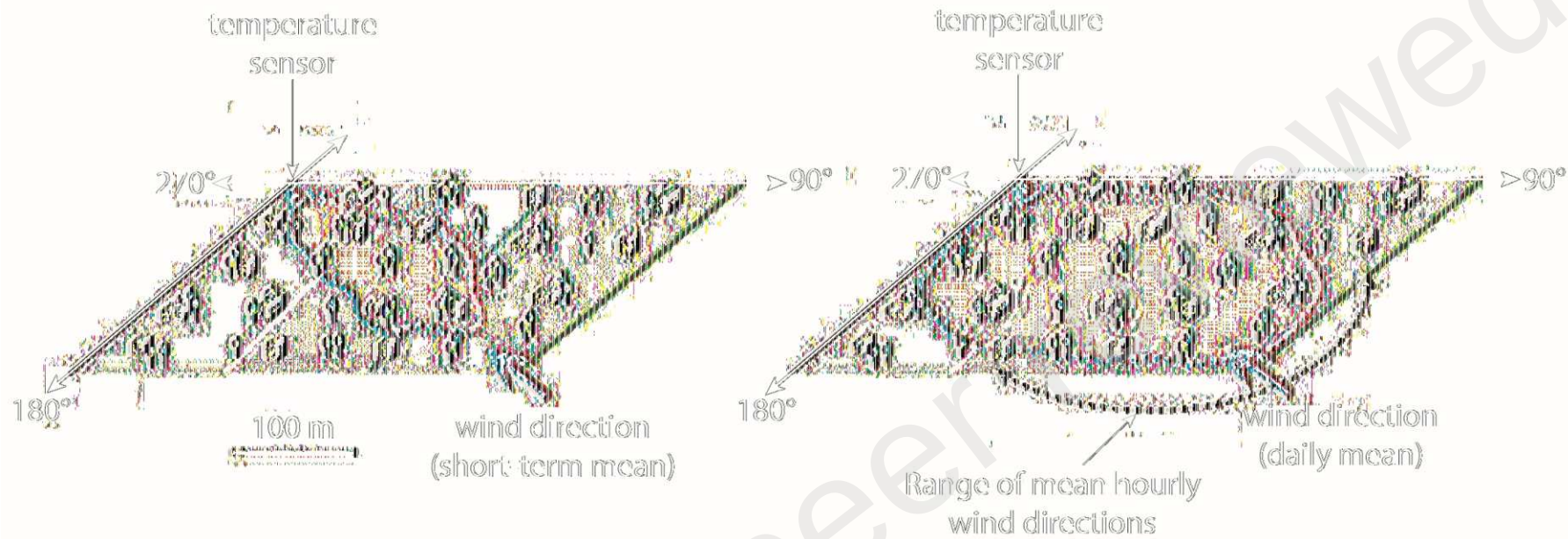


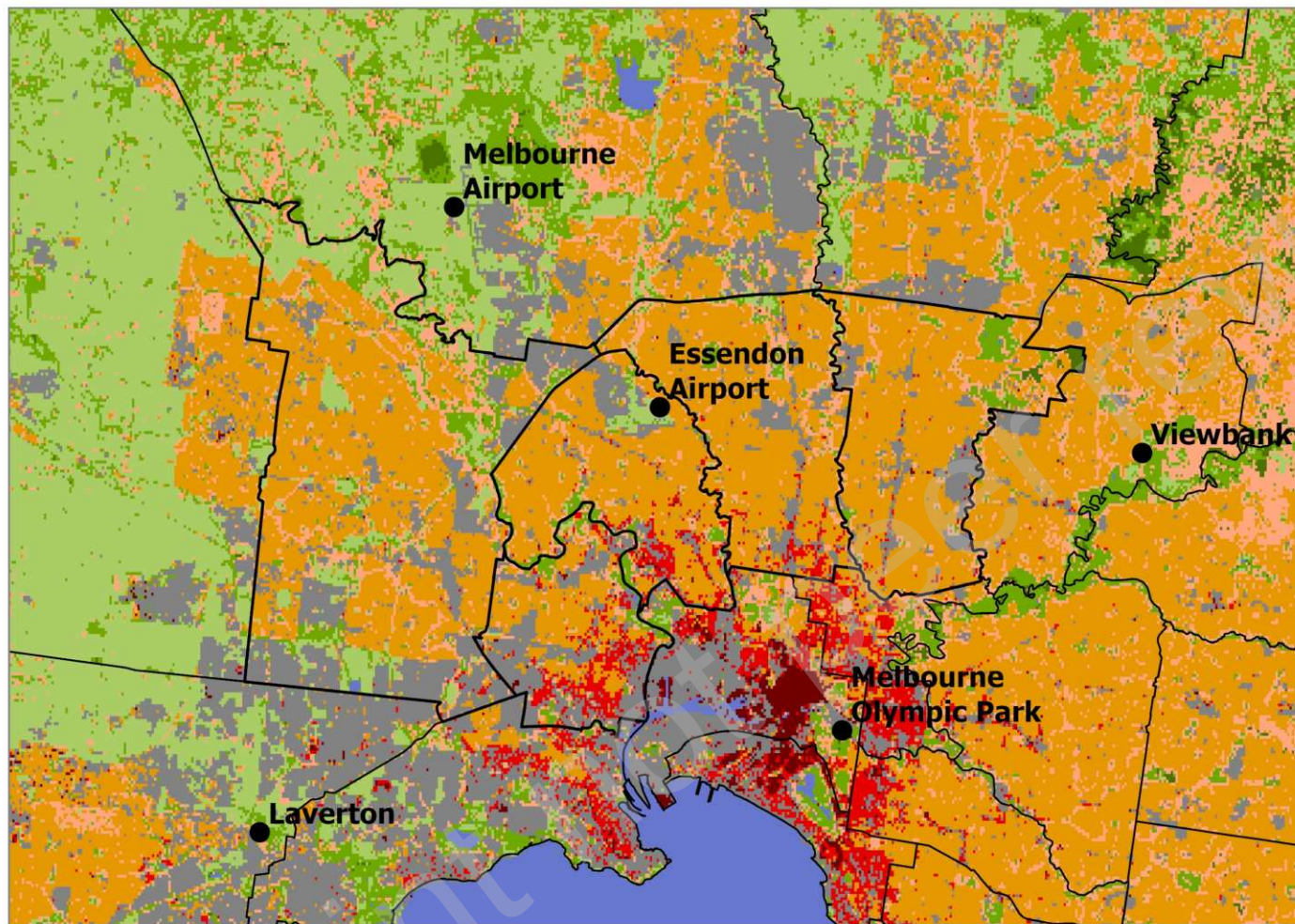
(3): LC/ 8. Large lowrise

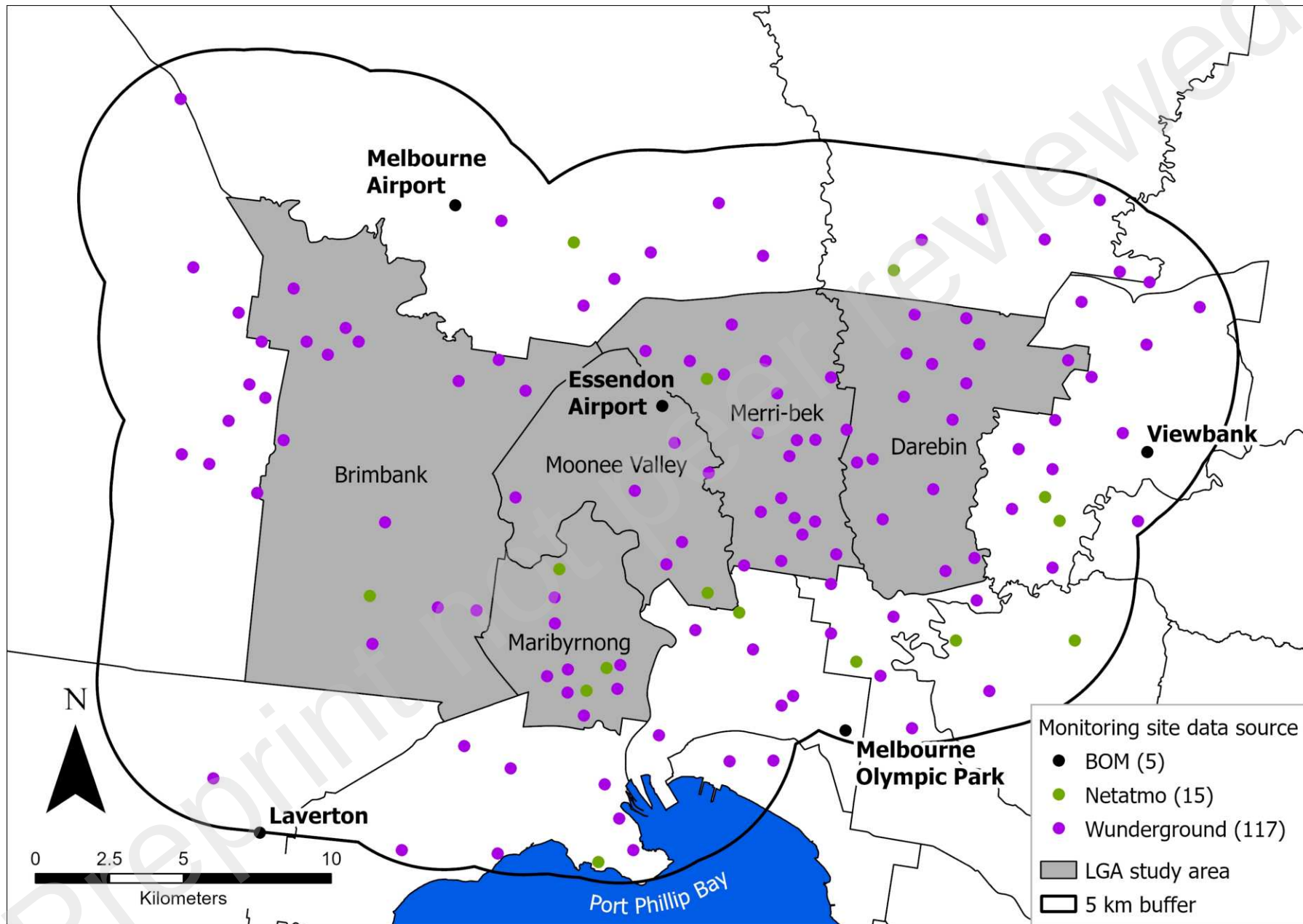


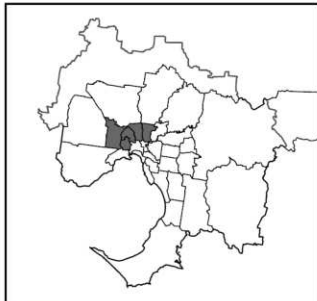
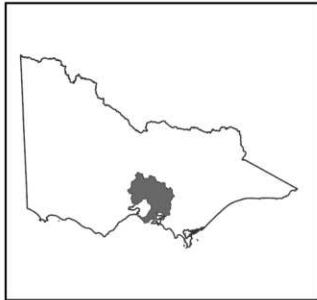


- Sensor
- 《 》 Thermal source area
- > Wind direction









-  LGA study area
-  BOM stations
-  LGA boundaries

